EEBus UC Technical Specification

Time of Use Tariff

Version 1.0.0

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1 Scope of the document

This document describes the Use Case "Time of Use Tariff" (short-name: TOUT). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification]	EEBus_SPINE_TS_ProtocolSpecification.pdf
[ResourceSpecification]	EEBus_SPINE_TS_ResourceSpecification.pdf
[SHIP]	EEBus_SHIP_TS_Specification_v1.0.1.pdf

1.1.2 Normative references

[RFC2119]	IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor
An Actor models a role within a Use Case definition (e.g. an energy manager or a heat pump).

CEM
Abbreviation for Customer Energy Manager. The CEM is an energy manager located at the user's building or in a cloud application. The energy manager enables energy-optimized operation of the connected devices by harmonizing energy demand and availability.

DSO
Abbreviation for "Distribution System Operator"

Energy Broker
A logical or physical unit at the customer's ESP

ESP
Abbreviation for "Energy Service Provider"

EV
Abbreviation for "Electric Vehicle"

PODF
Short name of the Use Case "Power Demand Forecast"

POEN
Short name of the Use Case "Power Envelope"

PoE
Abbreviation for "Price of Energy"

Scenario
Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or optional.

Specialization
Reusable data collection for a specific functionality.

SPINE
Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative e.V.

TF
Abbreviation for "Transmission Fee"

TOUT
Time of Use Tariff (short name of this Use Case)

UC
Abbreviation for "Use Case"

1.3 Requirements

1.3.1 Requirements wording

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT
- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

1.3.2 Mapping of High-Level requirements

Within the High-Level Use Case description, the following abbreviation is used:

[TOUT-xyz]

e.g.: [TOUT-007]

The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique number xyz. These requirements are referenced throughout the technical solution to show how each High-Level requirement is realized in the technical part.

2 High-Level description

2.1 Introduction

The electricity grid is more and more challenged by energy consuming appliances such as electric vehicles (EVs), heat pumps, etc. on the one hand and transition to regenerative energy production on the other hand. To enable this, energy demand and energy production need to be harmonized. Energy Service Providers (ESPs) or Distribution System Operators (DSOs) need to time-shift the energy demand of consumers to manage over and underload scenarios in the distribution grid. As many appliances are not capable or willing to be shifted directly, a mechanism to trigger the appliances to shift their demand themselves is needed.

This Use Case provides an incentive-based mechanism for load shifting. For this, the ESP or the DSO may send incentive tables to the customer energy manager (CEM) of buildings. The CEM may manage the connected appliances within their flexibility (which depends on the capabilities of the appliances) to optimize the power consumption or production according to the communicated incentives, e.g., price, CO_2 emission, etc.

Within this Use Case the following Actors are used:

- "Energy Broker" as optimization instance at the customer's ESP
- "Transmission Broker" for the DSO's or the ESP's role in this Use Case (depending on the overall system setup)
- "CEM" for the energy manager at the customer's building.

Depending on the contracts of the customer with the ESP and the DSO and the regulations in the country they are located, either both Actors send individual incentive tables to the CEM or only the ESP (see section 2.2.1).

Each incentive table with a tariff can have multiple tariff levels (called "tiers") which have one or more incentives (e.g., absolute price, CO_2 emission, etc.). Based on the incentives, the CEM may calculate where to plan the consumption and production cycles of its managed appliances (see section 2.2.4). At least, the absolute price SHALL be included in the incentive table.

Within this Use Case, a time-based incentive information approach is used to model incentive tables. Subsequently, these are called "incentive-based incentive tables".

The costs/profits for the energy consumed/produced by the building are composed of several individual items, depending on the country's regulations, the contract, etc. The two main items are the price of energy (PoE) that is billed by the ESP and the transmission fee (TF) that is billed by the DSO (see section 2.2.5). Depending on the overall system setup, either the Energy Broker combines both items within its incentive table ("total price" = PoE + TF), or they are transmitted separately by Energy Broker and Transmission Broker to the CEM.

Note: The costs or profits being included in an incentive table do not necessarily depend on the energy direction, the incentive table is defined for (e.g., there may also be profits in an incentive table for consuming energy to stabilize the grid, i.e., if too much energy is available and needs to be consumed).

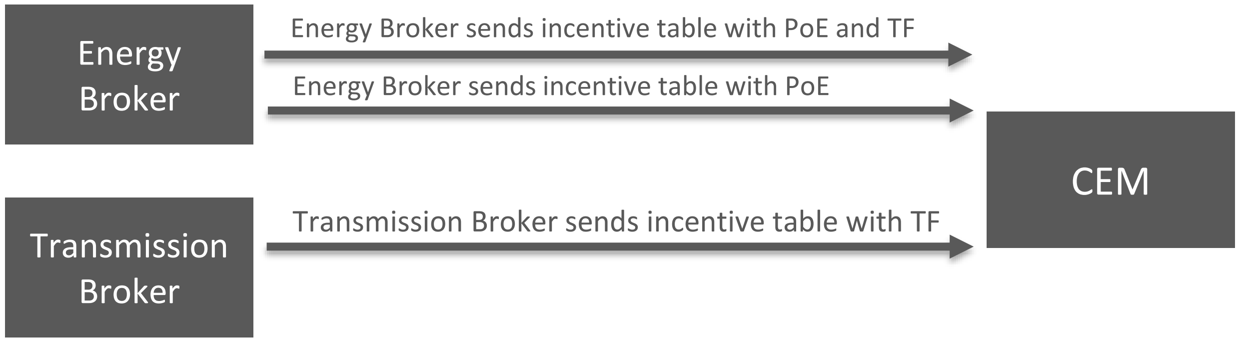

Figure 1: High-Level Use Case functionality overview

2.2 Detailed background information and rules

2.2.1 Information and rules regarding the Use Case's Actors

As the previous section introduces, there are three Actors within this Use Case (Energy Broker, Transmission Broker and CEM). The Actor CEM is always needed for the Use Case's functionality. But whether Energy Broker, Transmission Broker or both are part of the UC depends on the overall electricity grid setup, the country's regulations and the partners the customer chose for the electricity supply. The CEM SHALL be able to receive incentive tables from both Actors. Within this Use Case, the Actor CEM SHALL accept incentive tables from at most one Actor Energy Broker as well as at most one Actor Transmission Broker.

2.2.2 Details and rules on incentive tables

An incentive table describes incentives over time. Each incentive can depend on the consumed or produced power. This dependency is expressed with so-called tiers (see section 2.2.4). The time is split into slots. The following figure gives an example for an incentive table:

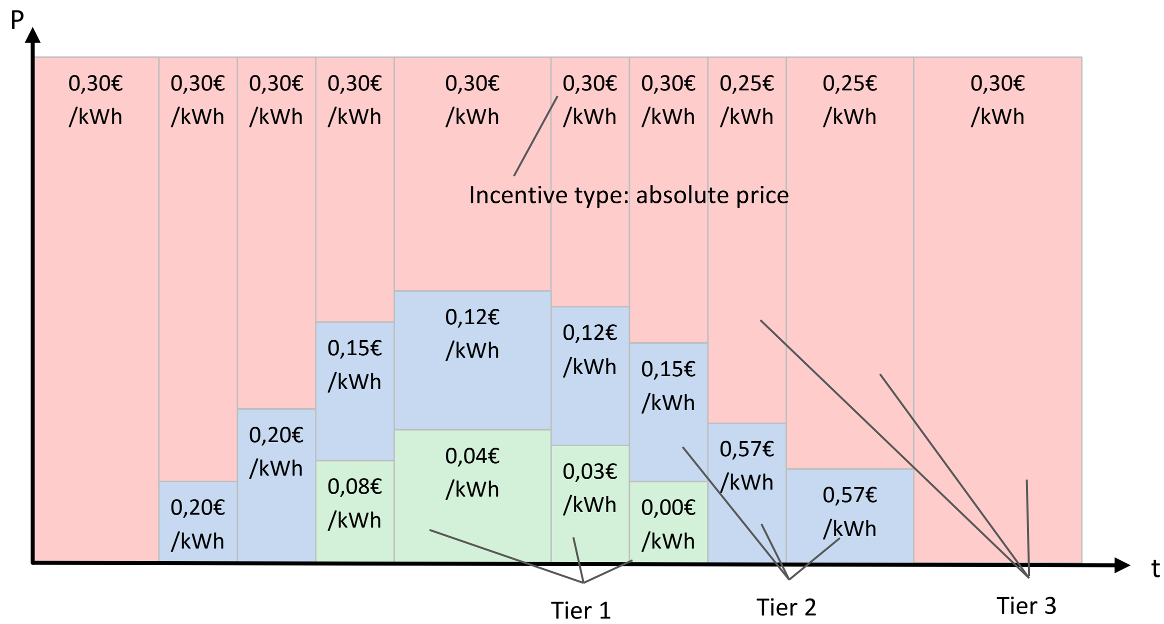

Figure 2: Incentive table example for consumption

The incentive table example in Figure 2 includes up to three different tiers and a total of 10 slots.

In this Use Case so-called "unique tiers" are used (in contrast to "stacked tiers" used in some other Use Cases). As can be seen in Figure 3 and Figure 4, the "price/kWh" is determined completely by the tier's power range in which the power P resides at a given time. The price is then calculated as follows:

$$Cost_{total}/h = P * price/kWh$$

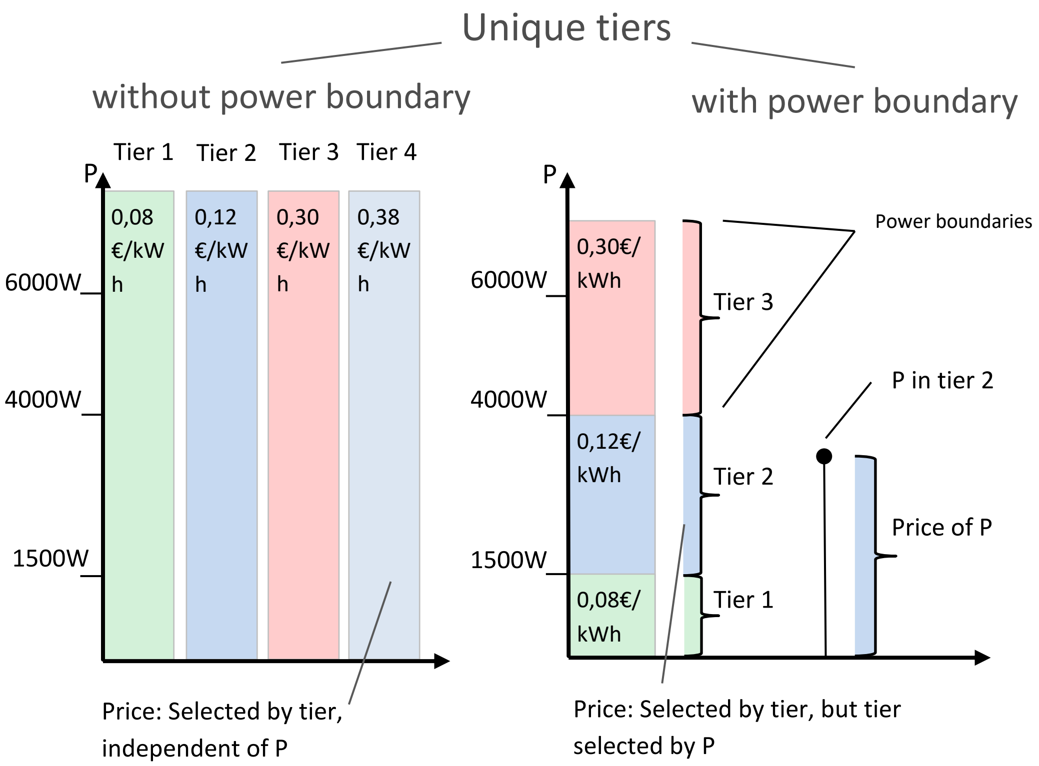

Figure 3: Unique tiers example for consumption

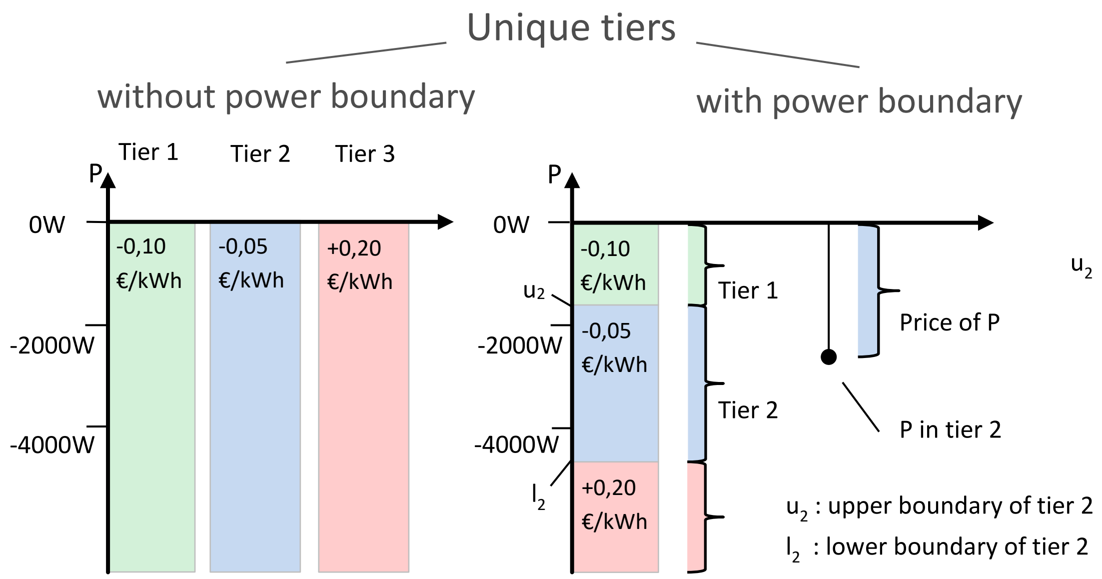

Figure 4: Unique tiers example for production

The same applies not only to monetary costs, but also to environmental costs. It should be noted that the values between one tier and the next lower or higher tier should not differ significantly, as (e.g.) CO2 emissions (per kWh) do not change significantly just because the tier limit is exceeded.

For power consumption incentive tables the power boundaries of the tiers SHALL be greater or equal to zero (W) ([TOUT-001a]) with the upper boundary greater than the lower boundary ([TOUT-002a]).

For power production incentive tables the power boundaries of the tiers SHALL be lower or equal to zero (W) ([TOUT-001b]) with the upper boundary greater than the lower boundary ([TOUT-002b]).

There SHALL NOT be gaps between power boundaries of adjacent tiers ([TOUT-002c]).

The value range of the boundaries SHALL be large enough so that one tier is always valid, no matter how much energy the CEM consumes or produces (e.g. the value of the last upper boundary value of the consumption direction could be 10^{10}) ([TOUT-002d]).

Note: This Use Case does not prescribe a particular order of the incentive magnitudes within a slot. E.g., with regard to price-based incentives a "lower" tier may be more expensive than an "upper" tier.

The Actor CEM stores incentive tables received by the Actors Energy Broker and Transmission Broker. The Actor CEM SHALL support incentive tables with up to 3 tiers and it SHOULD support incentive tables with up to 5 tiers. The Actor CEM SHALL express the number of supported tiers ([TOUT-003]).

Each tier MAY include a human readable description ([TOUT-004]). The human readable description can be freely chosen by the Actor sending an incentive table. Note: The value range of the human readable description is not standardized in any way, thus the meaning of the human readable description should only be interpreted by human interaction and not by a device itself.

The examples above only show price-based incentives. In general, each tier can comprise of up to three distinct incentive types ([TOUT-005]). The following incentive types are permitted:

- Absolute price per energy (unit currency/Wh) ([TOUT-006])
- Renewable energy percentage ([TOUT-007])
- CO_2 emission per energy (unit kg/Wh) ([TOUT-008])

Each tier SHALL contain at least the absolute price ([TOUT-006]).

Note: Negative values for absolute prices are permitted and SHALL be interpreted as profits.

The set of incentive types per tier SHALL be the same throughout the whole incentive table, i.e. across all tiers and all slots. This means that if one tier of one slot contains an incentive for an absolute price in (e.g.) €/Wh, and also an incentive for renewable energy percentage, then all other tiers in all other slots of the incentive table SHALL have these two incentive types as well.

2.2.3 Time rules for incentive tables

An incentive table SHALL NOT begin later than "now". This means that an incentive table that is submitted at time "x" is only valid if the table states an incentive for a slot that covers time "x".

The start time and end time of each slot SHALL be expressed as UTC values ([TOUT-009]). An incentive table SHALL NOT have time gaps between two slots ([TOUT-010]). This means the end time of a slot SHALL be determined by the start time of its subsequent slot. The last slot of an incentive table SHALL state its end time explicitly as UTC value.

An incentive table SHALL NOT be longer than 7 days (168 hours).

The following additional rules regarding the time span of an incentive table apply:

1. The incentive table SHALL have slots until time "x" + 24 hours at least.
2. The incentive table SHOULD have slots until time "x" + 48 hours.

Note (informative): An incentive table that covers a long time period may have a higher time resolution (i.e. more slots) for the near future, while slots of the distant future have a coarser time resolution. In general, the time resolution should be chosen to express significant changes in the tier (e.g. the power level or an incentive of a tier changes). E.g. if the tiers do not change over the time span of the incentive table, only a single time slot is needed.

2.2.4 Details on tiers (tariff levels) and incentives

Tiers (tariff levels) are used to model different levels within the tariff. There are several possible triggers when the tiers are switched:

- Time-based: At certain points in time the active tier switches to another one.
- Power-based: When more (or less) power is used the active tier switches.
- Energy-based (not considered within this Use case): When more energy was consumed since a specific point in time (e.g. the first day of the month) the active tier switches.

Switching tiers may result in higher (or even lower) "costs".

Incentives quantify the "costs" of energy. The most likely costs are monetary costs, but there can also be other incentives like CO_2 emission or renewable energy percentage.

The incentives are always bound to a specific tier. When the tier switches, the according incentives become valid.

2.2.5 Details on price of energy (PoE) and transmission fee (TF)

Two different items of the overall costs or profits of energy consumption or production are used within this Use Case:

- Price of energy (PoE)
- Transmission fee (TF)

The **price of energy** represents the costs or billed charged by the Energy Broker. The amount of the PoE for the monetary costs depends (among others) on the price the Energy Broker has paid at the electricity stock market, the consumption/production of the other buildings the Energy Broker has contracts with, etc. The type and method of electricity generation is responsible for the environmental costs (e.g. the percentage of wind or solar power). While it is mandatory for the Energy Broker to provide the monetary costs, the environmental costs are optional.

The **transmission fee** represents the costs the Transmission Broker has for transmitting the energy to the building. As more and more regenerative energy is produced decentralised, the challenges for the Transmission Broker to keep the electricity grid in a stable state are growing and thus the need for the buildings to shift their energy consumption and production rises. For the transmission fee, only monetary costs can be determined.

Note: Within this Use Case all monetary costs or profits are denoted as gross prices.

Note: Within this Use Case, monetary costs are denoted as positive value, profits as negative value.

Note: Within this Use Case, environmental costs can only be determined for consumption direction. Hence, they are always positive.

2.3 Actors

2.3.1 Energy Broker

The Actor Energy Broker submits tariff information to the building. The tariff may include the transmission fee and the price of energy (PoE+TF) or only the latter (PoE) (see section 2.2).

2.3.2 Transmission Broker

The Actor Transmission Broker sends tariff information into the building including the transmission fee only. This incentive is used by the Transmission Broker to motivate a CEM to shift the building's energy demand in times of lower costs that typically correlates with more available energy in the electricity grid.

2.3.3 CEM

The Actor CEM manages the devices in the building in order to shift the load to times of lower costs (e.g. monetary costs, CO_2 emission, etc.).

2.4 Scenarios

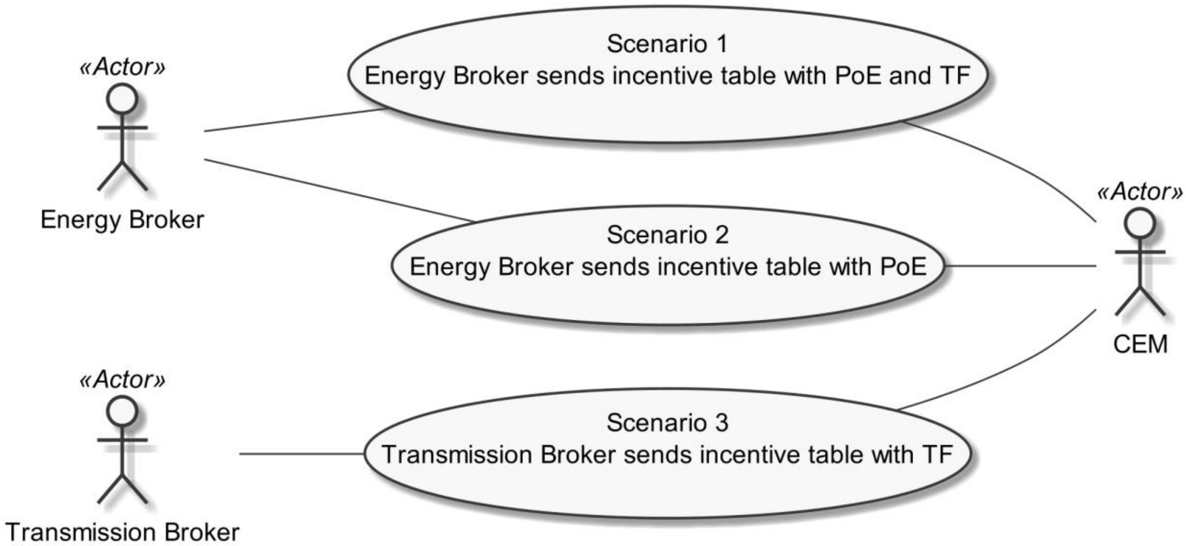

Figure 5: Scenario overview

Note: Figure 5 shows Scenarios for different configurations, depending on the contract between the Actors. For Scenario 1, the Actors Energy Broker and CEM exchange incentive tables with PoE and TF. As long as the respective contract between the Actors is valid, Scenarios 2 and 3 will not apply. This means that the Actor CEM can reject incentive tables that belong to Scenario 2 or 3. If, on the other hand, the Actor CEM is configured for Scenarios 2 and 3 (due to a proper contract), it can reject incentive tables that belong to Scenario 1. This means that Scenario 1 is not intended to be executed unconditionally in parallel to Scenarios 2 and 3. Whether the Actor CEM is able to alter its Scenario-configuration dynamically on its own or whether it provides a configuration option to the user is not described in this Use Case. Independent from the current configuration, the Actor CEM must generally be able to support the Scenarios as indicated in Table 1.

Note: If Scenario 2 and 3 are used (hence, an Energy Broker sends the incentive tables for the PoE and a Transmission Broker sends the incentive tables for the TF, both Actors should use the same combination of incentive tables for consumption and production (e.g., both, Energy Broker and Transmission Broker, send incentive tables for consumption and production, or, both send only incentive tables for consumption).

Scenario Number	Scenario Name	Energy Broker	Transmission Broker	CEM
1	Energy Broker sends incentive table with PoE and TF*[1]	M	-	M
2	Energy Broker sends incentive table with PoE*[2]	M	-	M
3	Transmission Broker sends incentive table with TF*[3]	-	M	M

Table 1: Scenario implementation requirements for Actors

*[1]: This Scenario SHALL only be used, if no Actor Transmission Broker is enabled to submit incentive tables to the CEM.

*[2]: This Scenario SHOULD only be used, if an Actor Transmission Broker is enabled and capable to submit incentive tables with the transmission fee to the CEM. In case where this Scenario is used without any Transmission Broker providing incentive tables with the transmission fee, the Actor CEM will not receive all costs (or profits) related to energy consumption or production of the building.

*[3]: This Scenario SHALL only be used by the Actor Transmission Broker, if it is enabled and capable to submit incentive tables with the transmission fee to the CEM.

2.4.1 Scenario 1 - Energy Broker sends incentive table with PoE and TF

2.4.1.1 Description

With this Scenario, the Actor Energy Broker sends incentive-based incentive tables with the price of energy and the transmission fee to the Actor CEM.

There can be different incentive tables for the consumption of power and the production. Whether both incentive tables are supported by the CEM depends on the CEM's capability itself as well as the connected appliances within the building and their capability of producing energy.

The Actor CEM SHALL always permit that the Actor Energy Broker sends an update for a supported incentive table ([TOUT-015]).

The Actor CEM MAY request towards Actor Energy Broker to get a modified or new incentive table ([TOUT-016]).

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Incentive-based incentive table for energy consumption with price of energy and transmission fee	Information about the incentive values per time-period for the price of energy and transmission fee for the consumption of energy.	M	[TOUT-011]
Incentive-based incentive table for energy production with price of energy and transmission fee	Information about the incentive values per time-period for the price of energy and transmission fee for the production of energy.	O*[1]	[TOUT-012]

Table 2: Scenario 1 - Energy Broker sends incentive table with PoE and TF - Data point list

*[1]: In general, the support for the Incentive Table for energy production is optional for the Actor CEM. The Incentive table for energy production SHALL only be supported by the Actor CEM if energy producing appliances are available in the building.

2.4.1.2 Conditions

Triggering Event:
The Actor Energy Broker wants to update one or more incentive tables on the Actor CEM.

Pre-condition:
The incentive table(s) of the Actor CEM are outdated.

Post-condition:
The Actor CEM has new incentive table(s).

2.4.2 Scenario 2 - Energy Broker sends incentive table with PoE

2.4.2.1 Description

With this Scenario, the Actor Energy Broker sends incentive-based incentive tables with the price of energy to the Actor CEM.

Note: Without the information of Scenario 3, the overall incentive values cannot be calculated completely.

There can be different incentive tables for the consumption of power and the production. Whether both incentive tables are supported by the CEM depends on the CEM's capability itself as well as the connected appliances within the building and their capability of producing energy.

The Actor CEM SHALL always permit that the Actor Energy Broker sends an update for a supported incentive table ([TOUT-025]).

The Actor CEM MAY request towards Actor Energy Broker to get a modified or new incentive table ([TOUT-026]).

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Incentive-based incentive table for energy consumption with price of energy	Information about the incentive values per time-period for the price of energy for the consumption of energy.	M	[TOUT-021]
Incentive-based incentive table for energy production with price of energy	Information about the incentive values per time-period for the price of energy for the production of energy.	O*[1]	[TOUT-022]

Table 3: Scenario 2 - Energy Broker sends incentive table with PoE - Data point list

*[1]: In general, the support for the Incentive Table for energy production is optional for the Actor CEM. The Incentive table for energy production SHALL only be supported by the Actor CEM if energy producing appliances are available in the building.

2.4.2.2 Conditions

Triggering Event:
The Actor Energy Broker wants to update one or more incentive tables on the Actor CEM.

Pre-condition:
The incentive table(s) of the Actor CEM are outdated.

Post-condition:
The Actor CEM has new incentive table(s).

2.4.3 Scenario 3 - Transmission Broker sends incentive table with TF

2.4.3.1 Description

With this Scenario, the Actor Transmission Broker sends incentive-based incentive tables with the transmission fee to the Actor CEM.

Note: Without the information of Scenario 2, the overall incentive values cannot be calculated completely.

There can be different incentive tables for the consumption of power and the production. Whether both incentive tables are supported by the CEM depends on the CEM's capability itself as well as the connected appliances within the building and their capability of producing energy.

The Actor CEM SHALL always permit that the Actor Transmission Broker sends an update for a supported incentive table ([TOUT-035]).

The Actor CEM MAY request towards Actor Transmission Broker to get a modified or new incentive table ([TOUT-036]).

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Incentive-based incentive table for energy consumption with transmission fee	Information about the incentive values per time-period for the transmission fee for the consumption of energy.	M	[TOUT-031]
Incentive-based incentive table for energy production with transmission fee	Information about the incentive values per time-period for the transmission fee for the production of energy.	O*[1]	[TOUT-032]

Table 4: Scenario 3 - Transmission Broker sends incentive table with TF - Data point list

*[1]: In general, the support for the Incentive Table for energy production is optional for the Actor CEM. The Incentive table for energy production SHALL only be supported by the Actor CEM if energy producing appliances are available in the building.

2.4.3.2 Conditions

Triggering Event:
The Actor Transmission Broker wants to update one or more incentives table on the Actor CEM.

Pre-condition:
The incentive table(s) of the Actor CEM are outdated.

Post-condition:
The Actor CEM has new incentive table(s).

2.5 Dependencies to other Use Cases

2.5.1 "Power Demand Forecast"

The Use Case "Power Demand Forecast" is used by the Actor CEM to provide information about its planned power consumption (or production, if available). Within a power demand forecast, the CEM can send multiple time slots, each with the estimated (planned) power consumption.

The Actor Energy Broker SHOULD support the Use Case "Power Demand Forecast".

The Actor Transmission Broker SHOULD support the Use Case "Power Demand Forecast".

The Actor CEM SHOULD support the Use Case "Power Demand Forecast".

The Actor names do not differ in both Use Cases.

2.5.2 "Power Envelope"

The Use Case "Power Envelope" enables the Actor Transmission Broker to limit the CEM's active power consumption or production. Each power envelope can have multiple time-slots with different values.

The Actor Transmission Broker SHOULD support the Use Case "Power Envelope".

The Actor CEM SHOULD support the Use Case "Power Envelope".

The Actor names do not differ in both Use Cases.

2.6 Further information and rules

2.6.1 Applied sign convention

This Use Case applies the "passive sign convention" ([TOUT-040]):

An electrical current is positive if the current is flowing into a building. In this case, the building consumes electrical power and the active power is greater than zero. An electrical current is negative if the current is flowing out of a building. In this case, the building produces electrical power and the active power is smaller than zero.

2.7 Assumptions and Prerequisites

None.

3 Technical SPINE solution

3.1 General rules and information

3.1.1 Underlying technology documents

This technical solution relies on the SPINE Resources Specification version 1.3.0 [ResourceSpecification].

For interoperable connectivity this technical solution relies on:

- SPINE Protocol Specification version 1.3.0 [ProtocolSpecification] as application protocol.
- SHIP Specification version 1.0.1 [SHIP] as transport protocol.

3.1.2 Use Case discovery rules

Use Case discovery SHALL be supported by each Actor and the following rules SHALL apply:

- The string content for the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseName" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be "timeOfUseTariff". The string content SHALL only be defined by this Use Case (regardless of the Use Case version).
- The string content of the Element "nodeManagementUseCaseData. useCaseInformation. actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to the according value stated within the corresponding Actor's section.
- An Actor A that is implemented to support this Use Case specification SHALL set the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion" within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- If an Actor A supports multiple versions of this Use Case with the same major version number, only the highest one SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple versions of this Use Case with the same major version number as supported by Actor A, the Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- If an Actor A supports multiple versions of this Use Case with different major version numbers, for each major version number only the highest version number SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions with a major version number not implemented by Actor A, it still might be possible to run the Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B as a valid partner for this Use Case.

3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

3.1.3.1 General presence indication definitions

Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

Abbreviation	Meaning	Link to requirement keywords

M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 5: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- M, O
- M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.3 Presence indications for "Content of Function..." tables

This section only defines rules for the server side.

Elements that are marked with "M" SHALL be supported by the server in case of readable as well as writeable data. In case of writeable data (marked with "M \W") the server does not need to set the Element, because the Element is set only by the client.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be provided by the server.
- "O" means that the data MAY be provided by the server.
- "F" means that the data SHALL NOT be provided by the server.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be supported. In other words: If the client writes the Element, the server SHOULD accept those messages and the contained Elements.
- "O" means that the data MAY be supported. In other words: If the client writes the Element, the server MAY accept those messages and the contained Elements.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "read" request: The according Element MAY be set in the reply.
- In case of a received "write" operation: The server MAY ignore the Element.
- In case of a "notify" operation to be created: The server MAY set the Element.

Note: The server will only accept write operations if the result fulfils the server Function requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY result in an error message.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- M, O
- M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.4 Cardinality indications on Elements and list items

A cardinality indication on an Element or list item expresses constraints on the number of occurrences of a given Element or data set. In this section we use "X" as representation for such an Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

1. X
   No cardinality indication.
2. X (a..b)
   This means "X" SHALL occur at least "a" times and at maximum "b" times.
3. X (a..)
   This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
4. X (..b)
   This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even zero occurrences are permissive).

Note: These rules apply only under consideration of presence indications and with regards to the given Scenario or Function definition for this Use Case.

The following table is an example to explain this for two different placements.

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
...	...		...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*#(1..)>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

Table 6: Example table for cardinality indications on Elements and list items

The field

xFeatureType. xListData. xData. (1..3)

introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2. The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3 times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

2: M \W

denotes that this data set is mandatory for Scenario 2.

The field

xSlot. (1..)

expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur more than one time.

For the expression "<*#(1..)>" of Element "xId" please see section 3.1.3.6.

The remaining fields do not have an explicit cardinality indication.

Note: Cardinality expressions are also used within placeholder expressions as defined in section 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted Elements or list items as they explicitly define how many different values for a given Identifier are required or permitted for a given Scenario.

3.1.3.5 Writability and changeability indication

In the same column where the presence indications are denoted, a mark is used to distinguish between writeable, changeable or readable Elements:

- Elements that are marked with "\W" are written by a client and SHALL be writeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\C" are changed by a client and SHALL be changeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\RW" are read and written by a client and SHALL be writeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are marked with "\RC" are read and changed by a client and SHALL be changeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are not marked are only read by a client and SHALL be provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.

"Writeable" means that the Element and its value may be written by a client. This includes the possibility to modify (if the Element is already present), create (if the Element is not present yet), and delete the Element. The server SHALL adjust its Function according to the received "write" operation (unless the server cannot accept the "write" operation according to section 3.1.3.3).

"Changeable" means that the Element's value may be changed by a client. If the Element is not present at the resource before, it probably **cannot** be created by the client via the "write" operation. In this case the server MAY decline such a message.

Note: "\W" includes "\C" already.

Note: Depending on the resource a client might need to request a proper binding before the server accepts a "write" operation.

3.1.3.6 "Value" placeholders

3.1.3.6.1 Introduction

Specializations may use placeholders to model relations between different Elements or even list items of different Functions. The main purpose is to declare which Identifier values relate to each other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder like "<x1>" can be used in "Value" columns to express the intended relations.

There are two styles to identify a placeholder that can be referenced:

1. <xM>
2. <xM#S>

where

1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
2. "M" is a (major) number like "1", "2", "15", etc.
3. "S" is a sub-number like "1", "7", "10", etc.

Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>", "<m22#3>". For a given placeholder, only one of the styles can be used.

In addition, there are also styles for placeholders that do not need to be referenced:

1. <*>
2. <*#S>

The second style is only used with so-called cardinality expressions.

3.1.3.6.2 Uniqueness of placeholders

A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case specification for a given set of its parent Identifier values. This shall be explained in a brief example:

We assume a list item with PRIMARY IDENTIFIER "pId". It also has a SUB IDENTIFIER "sId" with placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of pId. This means that "<s1>" of a list item with pId=1 can differ from "<s1>" of a list item with pId=2. But it also means that "<s1>" represents the same value in all list items with pId=1.

Note: Typically, parent Identifiers like "pId" will also be expressed with a placeholder like "<p5>", e.g. In this case, the uniqueness rule applies for "<p5>" likewise.

Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

3.1.3.6.3 Placeholder expressions with cardinalities

For some Identifiers, more than one placeholder is needed. Several notations are used for this purpose, which make use of cardinality expressions. The general notation is as follows:

1. <xM#(a..b)>

This is equivalent to this explicit definition:

At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a" distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the respective Identifier.

Additionally, the following notations may occur:

2. <xM#(a..)>
   This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.
3. <xM#(..b)>
   This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

"<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to "b" distinct values.

Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*#(a..b)> etc.

Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given, it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

3.1.3.6.4 References to placeholders and relations

According to the styles for placeholders that can be referenced, an enumeration value "e" can refer to a particular placeholder:

1. e(-><xM>)
2. e(-><xM#S>)

This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

Example: A Specialization contains the Elements "mId" and "phase". "mId" is an Identifier with placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using this expression:

"a"(-><m2#1>)
"b"(-><m2#2>)
"c"(-><m2#3>)

This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to be used with <m2#2> and "c" with <m2#3>.

Similarly, a placeholder "yN" can refer to a particular placeholder:

3. <yN->xM>
4. <yN->xM#S>
5. <yN#T->xM>
6. <yN#T->xM#S>

where "T" is a sub-number of "yN".

It is also feasible to associate placeholders with cardinalities:

7. <yN#(a..b)->xM#(a..b)>

denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

In some cases, there is a need to express that multiple list items with similar values are feasible or required, but only particular combinations of these different data are permitted. The following example shows that several "fData" list items with different "phase" value are required, but that these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase" value combination { "a", "abc", "neutral" }. The permitted combinations are defined in a note below a table:

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	F. fListData. fData.		
2: M	zId	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	
		"abc"(-><z3#4>)	
		"neutral"(-><z3#5>)	

Table 7: Content of an example Specialization

Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or {<z3#1>, <z3#4>, <z3#5>}.

3.1.3.7 Rules for content of "Value" column

For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element to one or more particular values. This means that Elements with values deviating from the restriction (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered as if the Element is not set. If more than one particular value is permitted for an Element, the values are in a single line each.

If a presence indication is set for the value (in an additional column before the value), the following rules SHALL be applied:

- "M" means that the value SHALL be supported. This means the value needs to be set at a certain point in time (depending on the value rules) or for a certain Element within a list entry.
- "R" means that the value SHOULD be supported.
- "O" means that the value MAY be supported.

If all possible values of a given mandatory Element are optional or recommended and this Element is used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values of a given optional or recommended Element are optional or recommended, this Element MAY contain also other values, but then this Element SHALL NOT be considered as part of the respective Scenario.

M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

If no presence indication is set for the value, the following rules SHALL be applied:

- In case of Elements where the server may set or change an Element on its own (see section 3.1.3.5):
  - within the tables in the "Server data - Resources" sections:
    - the server SHALL support at least one of the listed values.
  - within the tables in the "Client data - Specializations" sections:
    - the client SHALL support all listed values.
- In case of Elements that are writeable or changeable (see section 3.1.3.5):
  - within the tables in the "Server data - Resources" sections:
    - the server SHALL support all listed values.
  - within the tables in the "Client data - Specializations" sections:
    - the client SHALL support at least one of the listed values.

Depending on the Element, different values may be used during runtime. If this is the case, those rules are described within the value rules.

If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value MAY deviate from this, e.g. "(1024)".

3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of Specialization..." tables

Within the "Client data - Specializations" sections each Specialization is described in an own sub-section with the name "Specialization "<name of the Specialization>"" (e.g. "Specialization "Measurement_GridFeedInEnergy""). It contains only one table that includes all Elements needed for this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey background colour. This row contains the following parts:

<Feature Type>. <Function>.[<list entry instance name>.]

The <list entry instance name> is only included if the <Function> is a list-based Function. An example could be:

DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData.
deviceConfigurationKeyValueDescriptionData.

In the following rows, only the names of the Elements are stated, without the prefix described above.

Within the "Server data - Resources" sections each Feature Type is described in an own sub-section with the name "Feature Type "<name of the Feature Type>"" (e.g. "Feature Type "Measurement""). It contains sub-sections for each Function named "Function "<name of the Function>"" (e.g. "Function "measurementListData""). These sections contain one table with all Elements needed for this resource. The list entries are mentioned in a continuous row, highlighted with grey background colour. This row contains the following parts:

<Feature Type>. <Function>.[<list entry instance name>.]

The <list entry instance name> is only included if the <Function> is a list-based Function. An example could be:

Measurement. measurementDescriptionListData. measurementDescriptionData.

In the following rows, only the names of the Elements are stated, without the prefix described above.

For both kinds of tables, the following applies:

- Parent Elements are marked with a dot at the end of the name:
  <parent Element>.
  E.g.:
  value.
- If there are sub-Elements, they are described in own rows with the name of the parent Element as prefix, separated by a dot and a blank space:

<parent Element>. <sub-Element>

E.g.:

value. number

3.1.4 Rules for "Feature Types and Functions..." tables

3.1.4.1 Presence indications for "Feature Types and Functions..." tables

The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 8: Presence indication of Feature Types and Functions support

If at least one Function of a Feature has the presence indication "M", it is mandatory to support the Feature.

3.1.4.2 Rules for "Possible operations" column

Within the "Feature Types and Functions..." tables the column "Possible operations" state whether the Function is read- or writeable (as defined in the detailed discovery mechanism, see [ProtocolSpecification]).

If the "partial" concept (also called "restricted function exchange") SHALL be supported, the following notation is used (separated for read and write access):

read (M). partial (M)
write (M). partial (M)

If the "partial" concept SHOULD be supported, the following notation is used:

read (M). partial (R)
write (M). partial (R)

If the "partial" concept MAY be supported, the following notation is used:

read (M). partial (O)
write (M). partial (O)

The server can decide whether a notification is submitted complete or partial (as described in [ProtocolSpecification]) if not defined differently within this Use Case Specification.

3.1.5 "Actor ... overview" diagram rules

Within the "Actor [...] overview" diagrams in the "Actors" sub-sections the complete functionality of this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in Table 1. The Actor MAY have more functionality implemented than needed for this Use Case.

For the following Actor overview example, a brief description of the graphical symbols will be described.

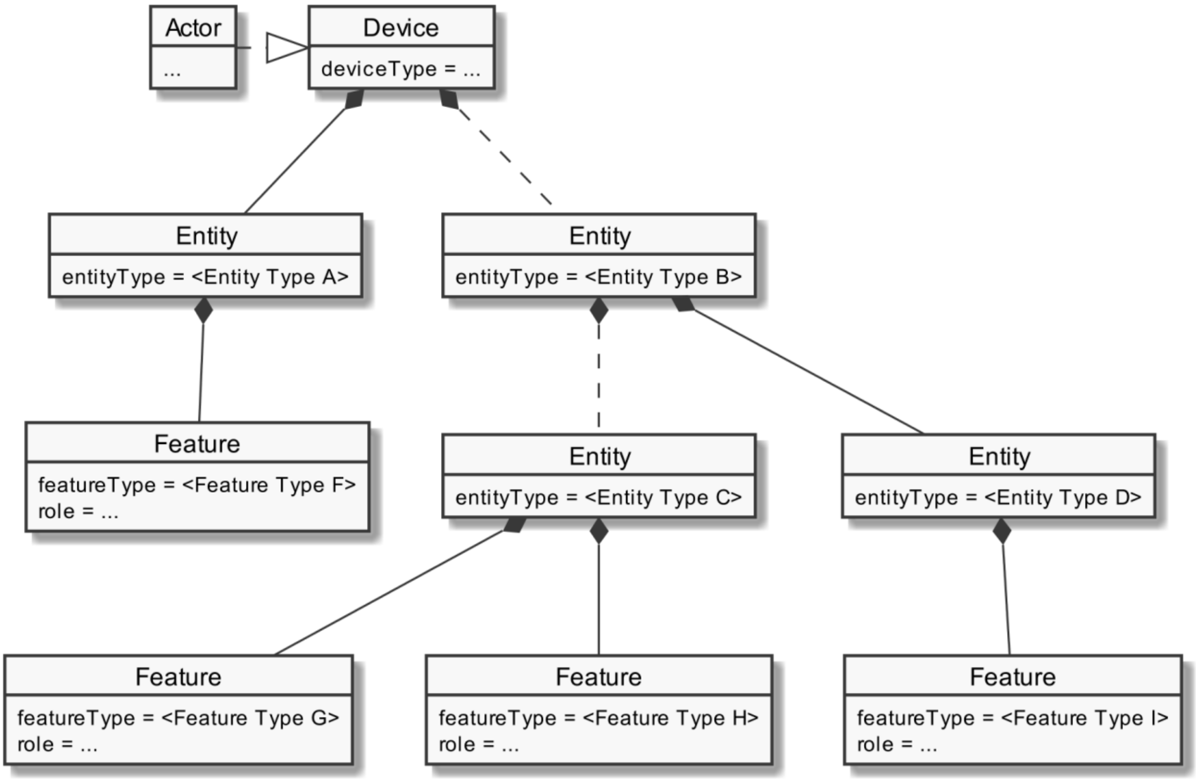

Figure 6: Actor overview example

The solid lines in the figure represent an immediate parent-childhood relation: The Entity with "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

The dashed lines in the figure express that there MAY be additional Entities between the shown Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

3.1.6 Specializations

Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often, data from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a Specialization defines a dataset that may encompass multiple related Functions from one or more different Features.

As different Use Cases sometimes share similar requirements, Specializations are also important from a re-usability perspective. This approach is used to improve consistency across Use Cases and avoid multiple variances of basically the same dataset. This is especially important in the case when an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two different Use Cases, both Use Cases could define slightly different datasets. In this case, the server as well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Scaled number rules]", e.g.

[Scaled number rules]:

The sub-Elements "number" and "scale" represent a value according to the following formula: number * 10^{scale}

3.1.8.2 Rules regarding the usage of time-related information

Timestamps used within this Use Case SHALL be presented as absolute times. This means relative times SHALL NOT be used for timestamps.

Note: This means the Actors need to have a common understanding of absolute times. It is strongly recommended that each Actor is "sufficiently synchronised" with the official "Coordinated Universal Time" (UTC).

3.1.8.3 Applied sign convention

For the applied sign convention, please see section 2.6.1.

3.2 Actors

3.2.1 Energy Broker

3.2.1.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "EnergyBroker" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Energy Broker's resource hierarchy.

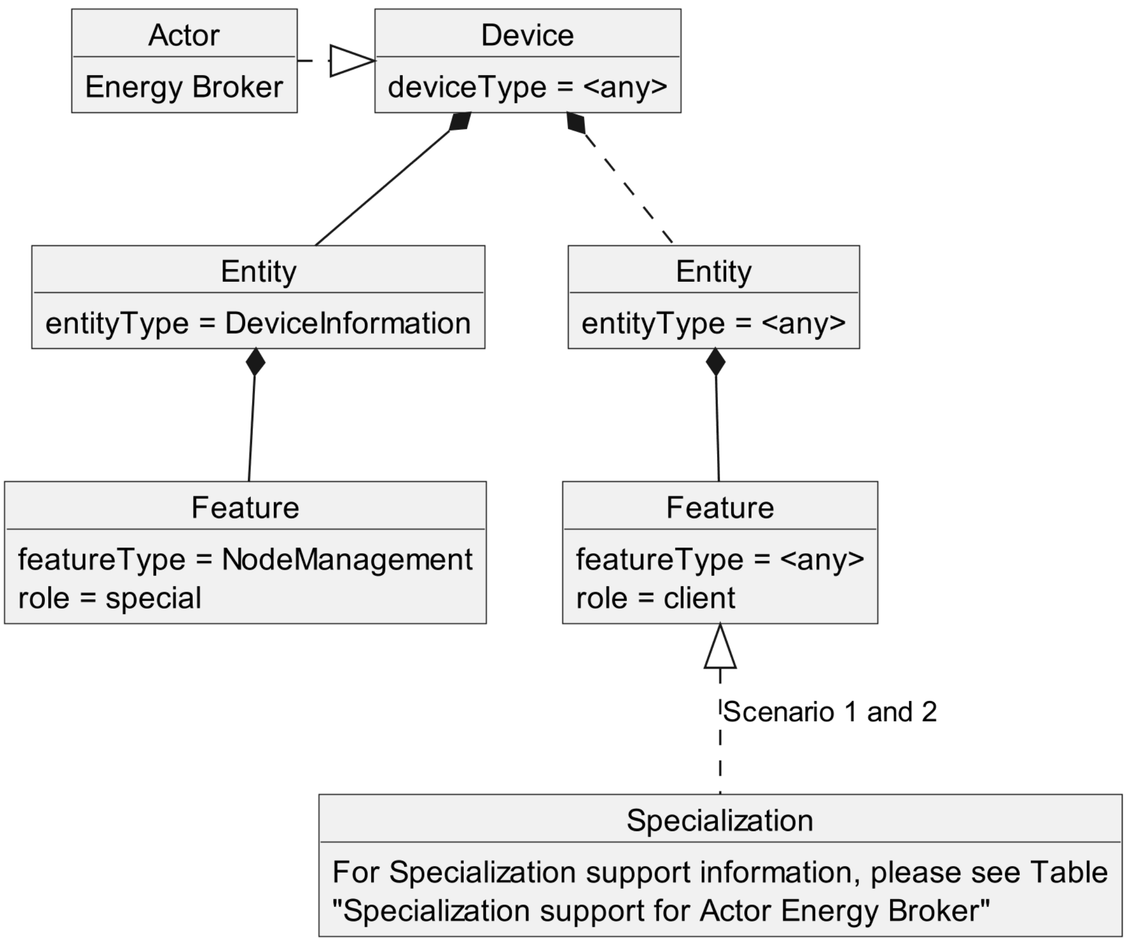

Figure 7: Actor "Energy Broker" overview

The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Described in table
IncentiveTable_EnergyConsumptionWithPoEandTF	1	Table 10
IncentiveTable_EnergyProductionWithPoEandTF	1	Table 11
IncentiveTable_EnergyConsumptionWithPoE	2	Table 12
IncentiveTable_EnergyProductionWithPoE	2	Table 13

Table 9: Specialization support for Actor Energy Broker

3.2.1.2 Server data - Resources

As this Actor has only client functionality, no resources are described within this section.

3.2.1.3 Client data - Specializations

3.2.1.3.1 Topic "IncentiveTable"

3.2.1.3.1.1 Specialization "IncentiveTable_EnergyConsumptionWithPoEandTF"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.		
1: M	tariffDescription.		[TOUT-011]
1: M	tariffDescription. tariffId	<tf1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

1: M	tariffDescription. tariffWriteable	"true"	[TOUT-015]
1: M	tariffDescription. updateRequired	"true" [TOUT-016]	With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
		"false"	
1: M	tariffDescription. scopeType	"incentiveTableEnConsWithPoETF"	
1: M	tariffDescription. slotIdSupport	"true"	
1: M \W	tier.		
1: M \W	tier. tierDescription.		
1: M \W	tier. tierDescription. tierId	<tr1#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
1: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
1: R \W	tier. tierDescription. label		[TOUT-004]
1: O \W	tier. tierDescription. description		[TOUT-004]

1: M \W	tier. boundaryDescription.			
1: M \W	tier. boundaryDescription. boundaryId	<by1#(1..1)>		SHALL be set as SUB IDENTIFIER of tierId.
1: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"		The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
1: M \W	tier. boundaryDescription. boundaryUnit	"W"		If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
1: M \W	tier. incentiveDescription.			
1: M \W	tier. incentiveDescription. incentiveId	<ie1#(1..3)>		SHALL be set as SUB IDENTIFIER of tierId.
1: M \W	tier. incentiveDescription. incentiveType	1: M \W	"absoluteCost"	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as charge. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
		1: O \W	"renewableEnergyPercentage"	[TOUT-007] The "unit" Element of the incentive SHALL be set to "pct" (percentage). The value SHALL be in the range from 0% to 100%.
		1: O \W	"co2Emission"	[TOUT-008] The "unit" Element of the incentive SHALL be set to "kg/Wh". Only positive values SHALL be used.
1: O \W, M \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
1: O \W, M \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
1: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.			
1: M	tariff.			
1: M	tariff. tariffId	<tf1#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
1: M	tariffConstraints.			

1: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5	[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
1: M	tariffConstraints. maxBoundariesPerTier	≥1	If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
1: M	tariffConstraints. maxIncentivesPerTier	≥1	[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
1: M	incentiveSlotConstraints.		
1: M	incentiveSlotConstraints. slotCountMax		If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
1: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
1: M \W	tariff.		
1: M \W	tariff. tariffId	<tf1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M \W	incentiveSlot.		
1: M \W	incentiveSlot. timeInterval.		[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
1: M \W	incentiveSlot. timeInterval. timeSlotId	<ti1#(1..)>	SUB IDENTIFIER of tariffId.
1: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
1: M \W	incentiveSlot. timeInterval. startTime. dateTime		
1: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
1: M \W	incentiveSlot. timeInterval. endTime. dateTime		
1: M \W	incentiveSlot. tier.		
1: M \W	incentiveSlot. tier. tier.		
1: M \W	incentiveSlot. tier. tier. tierId	<tr1#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
1: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary

			range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
1: M \W	incentiveSlot. tier. boundary. boundaryId	<by1#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	≥0	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	> incentiveSlot. tier. boundary. lowerBoundaryValue	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
1: M \W	incentiveSlot. tier. incentive.		
1: M \W	incentiveSlot. tier. incentive. incentiveId	<ie1#(1..3)>	SHALL be used as SUB IDENTIFIER of tierId.
1: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
1: M \W	incentiveSlot. tier. incentive. value. number		
1: M \W	incentiveSlot. tier. incentive. value. scale		

Table 10: Content of Specialization "IncentiveTable_EnergyConsumptionWithPoEandTF" at Actor Energy Broker

3.2.1.3.1.2 Specialization "IncentiveTable_EnergyProductionWithPoEandTF"

Scenario [{…}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.		
1: M	tariffDescription.		[TOUT-012]
1: M	tariffDescription. tariffId	<tf2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	tariffDescription. tariffWriteable	"true"	[TOUT-015]
1: M	tariffDescription. updateRequired	"true" [TOUT-016]	With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
		"false"	
1: M	tariffDescription. scopeType	"incentiveTableEnProdWithPoETF"	

1: M	tariffDescription. slotIdSupport	"true"	
1: M \W	tier.		
1: M \W	tier. tierDescription.		
1: M \W	tier. tierDescription. tierId	<tr2#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
1: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
1: R \W	tier. tierDescription. label		[TOUT-004]
1: O \W	tier. tierDescription. description		[TOUT-004]
1: M \W	tier. boundaryDescription.		
1: M \W	tier. boundaryDescription. boundaryId	<by2#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId.
1: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
1: M \W	tier. boundaryDescription. boundaryUnit	"W"	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
1: M \W	tier. incentiveDescription.		
1: M \W	tier. incentiveDescription. incentiveId	<ie2#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId.
1: M \W	tier. incentiveDescription. incentiveType	1: M \W | "absoluteCost"	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as charge. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
1: O \W, M \W	tier. incentiveDescription. currency		If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
1: O \W, M \W	tier. incentiveDescription. unit		If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
1: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.		
1: M	tariff.		

1: M	tariff. tariffId	<tf2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	tariffConstraints.		
1: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5	[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
1: M	tariffConstraints. maxBoundariesPerTier	≥1	If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
1: M	tariffConstraints. maxIncentivesPerTier	≥1	[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
1: M	incentiveSlotConstraints.		
1: M	incentiveSlotConstraints. slotCountMax		If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
1: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
1: M \W	tariff.		
1: M \W	tariff. tariffId	<tf2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M \W	incentiveSlot.		
1: M \W	incentiveSlot. timeInterval.		[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
1: M \W	incentiveSlot. timeInterval. timeSlotId	<ti2#(1..)>	SUB IDENTIFIER of tariffId.
1: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
1: M \W	incentiveSlot. timeInterval. startTime. dateTime		
1: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
1: M \W	incentiveSlot. timeInterval. endTime. dateTime		
1: M \W	incentiveSlot. tier.		
1: M \W	incentiveSlot. tier. tier.		

1: M \W	incentiveSlot. tier. tier. tierId	<tr2#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
1: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
1: M \W	incentiveSlot. tier. boundary. boundaryId	<by2#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	< incentiveSlot. tier. boundary. upperBoundaryValue	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	≤0	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
1: M \W	incentiveSlot. tier. incentive.		
1: M \W	incentiveSlot. tier. incentive. incentiveId	<ie2#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId.
1: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
1: M \W	incentiveSlot. tier. incentive. value. number		
1: M \W	incentiveSlot. tier. incentive. value. scale		

Table 11: Content of Specialization "IncentiveTable_EnergyProductionWithPoEandTF" at Actor Energy Broker

3.2.1.3.1.3 *Specialization "IncentiveTable_EnergyConsumptionWithPoE"*

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.		
2: M	tariffDescription.		[TOUT-021]
2: M	tariffDescription. tariffId	<tf3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	tariffDescription. tariffWriteable	"true"	[TOUT-025]
2: M	tariffDescription. updateRequired	"true" [TOUT-026]	With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
		"false"	
2: M	tariffDescription. scopeType	"incentiveTableEnConsWithPoE"	

<table>
<tr><td>2: M</td><td>tariffDescription. slotIdSupport</td><td colspan="2">"true"</td><td></td></tr>
<tr><td>2: M \W</td><td>tier.</td><td colspan="2"></td><td></td></tr>
<tr><td>2: M \W</td><td>tier. tierDescription.</td><td colspan="2"></td><td></td></tr>
<tr><td>2: M \W</td><td>tier. tierDescription. tierId</td><td colspan="2"><tr3#(1..5)></td><td>SHALL be set as SUB IDENTIFIER of tariffId.</td></tr>
<tr><td>2: M \W</td><td>tier. tierDescription. tierType</td><td colspan="2">"dynamicCost"</td><td>The tier has a cost incentive that MAY vary over time.</td></tr>
<tr><td>2: R \W</td><td>tier. tierDescription. label</td><td colspan="2"></td><td>[TOUT-004]</td></tr>
<tr><td>2: O \W</td><td>tier. tierDescription. description</td><td colspan="2"></td><td>[TOUT-004]</td></tr>
<tr><td>2: M \W</td><td>tier. boundaryDescription.</td><td colspan="2"></td><td></td></tr>
<tr><td>2: M \W</td><td>tier. boundaryDescription. boundaryId</td><td colspan="2"><by3#(1..1)></td><td>SHALL be set as SUB IDENTIFIER of tierId.</td></tr>
<tr><td>2: M \W</td><td>tier. boundaryDescription. boundaryType</td><td colspan="2">"powerBoundary"</td><td>The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.</td></tr>
<tr><td>2: M \W</td><td>tier. boundaryDescription. boundaryUnit</td><td colspan="2">"W"</td><td>If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.</td></tr>
<tr><td>2: M \W</td><td>tier. incentiveDescription.</td><td colspan="2"></td><td></td></tr>
<tr><td>2: M \W</td><td>tier. incentiveDescription. incentiveId</td><td colspan="2"><ie3#(1..3)></td><td>SHALL be set as SUB IDENTIFIER of tierId.</td></tr>
<tr><td rowspan="3">2: M \W</td><td rowspan="3">tier. incentiveDescription. incentiveType</td><td>2: M \W</td><td>"absoluteCost"</td><td>[TOUT-006]
Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as charge. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).</td></tr>
<tr><td>2: O \W</td><td>"renewableEnergyPercentage"</td><td>[TOUT-007]
The "unit" Element of the incentive SHALL be set to "pct" (percentage). The value SHALL be in the range from 0% to 100%.</td></tr>
<tr><td>2: O \W</td><td>"co2Emission"</td><td>[TOUT-008]
The "unit" Element of the incentive SHALL be set to "kg/Wh". Only positive values SHALL be used.</td></tr>
</table>

2: O \W, M \W	tier. incentiveDescription. currency		If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: O \W, M \W	tier. incentiveDescription. unit		If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.		
2: M	tariff.		
2: M	tariff. tariffId	<tf3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	tariffConstraints.		
2: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5	[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
2: M	tariffConstraints. maxBoundariesPerTier	≥1	If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
2: M	tariffConstraints. maxIncentivesPerTier	≥1	[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
2: M	incentiveSlotConstraints.		
2: M	incentiveSlotConstraints. slotCountMax		If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
2: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
2: M \W	tariff.		
2: M \W	tariff. tariffId	<tf3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M \W	incentiveSlot.		
2: M \W	incentiveSlot. timeInterval.		[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
2: M \W	incentiveSlot. timeInterval. timeSlotId	<ti3#(1..)>	SUB IDENTIFIER of tariffId.
2: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
2: M \W	incentiveSlot. timeInterval. startTime. dateTime		

2: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
2: M \W	incentiveSlot. timeInterval. endTime. dateTime		
2: M \W	incentiveSlot. tier.		
2: M \W	incentiveSlot. tier. tier.		
2: M \W	incentiveSlot. tier. tier. tierId	<tr3#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
2: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
2: M \W	incentiveSlot. tier. boundary. boundaryId	<by3#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	≥0	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	> incentiveSlot. tier. boundary. lowerBoundaryValue	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
2: M \W	incentiveSlot. tier. incentive.		
2: M \W	incentiveSlot. tier. incentive. incentiveId	<ie3#(1..3)>	SHALL be used as SUB IDENTIFIER of tierId.

2: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
2: M \W	incentiveSlot. tier. incentive. value. number		
2: M \W	incentiveSlot. tier. incentive. value. scale		

963 *Table 12: Content of Specialization "IncentiveTable_EnergyConsumptionWithPoE" at Actor Energy Broker*

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965 *3.2.1.3.1.4 Specialization "IncentiveTable_EnergyProductionWithPoE"*

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.		
2: M	tariffDescription.		[TOUT-022]
2: M	tariffDescription. tariffId	<tf4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	tariffDescription. tariffWriteable	"true"	[TOUT-025]
2: M	tariffDescription. updateRequired	"true" [TOUT-026]	With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the

		"false"	responsible client. The server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
2: M	tariffDescription. scopeType	"incentiveTableEnProdWithPoE"	
2: M	tariffDescription. slotIdSupport	"true"	
2: M \W	tier.		
2: M \W	tier. tierDescription.		
2: M \W	tier. tierDescription. tierId	<tr4#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
2: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
2: R \W	tier. tierDescription. label		[TOUT-004]
2: O \W	tier. tierDescription. description		[TOUT-004]
2: M \W	tier. boundaryDescription.		
2: M \W	tier. boundaryDescription. boundaryId	<by4#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId.
2: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
2: M \W	tier. boundaryDescription. boundaryUnit	"W"	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
2: M \W	tier. incentiveDescription.		
2: M \W	tier. incentiveDescription. incentiveId	<ie4#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId.

2: M \W	tier. incentiveDescription. incentiveType	2: M \W	"absoluteCost "	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as charge. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
2: O \W, M \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: O \W, M \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.			
2: M	tariff.			
2: M	tariff. tariffId	<tf4#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
2: M	tariffConstraints.			
2: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5		[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
2: M	tariffConstraints. maxBoundariesPerTier	≥1		If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
2: M	tariffConstraints. maxIncentivesPerTier	≥1		[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
2: M	incentiveSlotConstraints.			
2: M	incentiveSlotConstraints. slotCountMax			If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
2: M \W	IncentiveTable. incentiveTableData. incentiveTable.			
2: M \W	tariff.			
2: M \W	tariff. tariffId	<tf4#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
2: M \W	incentiveSlot.			
2: M \W	incentiveSlot. timeInterval.			[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.

2: M \W	incentiveSlot. timeInterval. timeSlotId	<ti4#(1..)>	SUB IDENTIFIER of tariffId.
2: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
2: M \W	incentiveSlot. timeInterval. startTime. dateTime		
2: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
2: M \W	incentiveSlot. timeInterval. endTime. dateTime		
2: M \W	incentiveSlot. tier.		
2: M \W	incentiveSlot. tier. tier.		
2: M \W	incentiveSlot. tier. tier. tierId	<tr4#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
2: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
2: M \W	incentiveSlot. tier. boundary. boundaryId	<by4#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	< incentiveSlot. tier. boundary. upperBoundaryValue	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	≤0	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		

2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
2: M \W	incentiveSlot. tier. incentive.		
2: M \W	incentiveSlot. tier. incentive. incentiveId	<ie4#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId.
2: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
2: M \W	incentiveSlot. tier. incentive. value. number		
2: M \W	incentiveSlot. tier. incentive. value. scale		

Table 13: Content of Specialization "IncentiveTable_EnergyProductionWithPoE" at Actor Energy Broker

3.2.2 Transmission Broker

3.2.2.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "TransmissionBroker" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Transmission Broker's resource hierarchy.

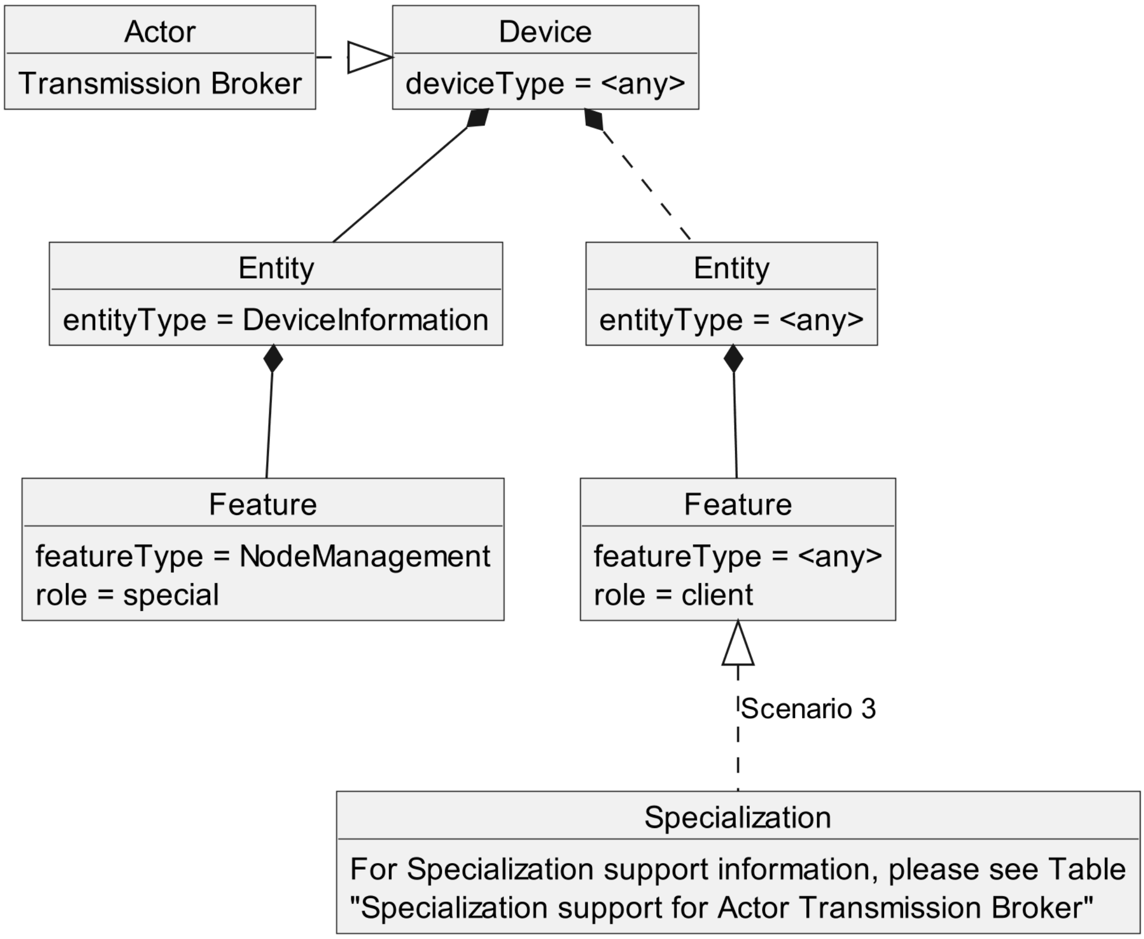

Figure 8: Actor "Transmission Broker" overview

The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Described in table
IncentiveTable_EnergyConsumptionWithTF	3	Table 15
IncentiveTable_EnergyProductionWithTF	3	Table 16

Table 14: Specialization support for Actor Transmission Broker

3.2.2.2 Server data - Resources

As this Actor has only client functionality, no resources are described within this section.

3.2.2.3 Client data - Specializations

3.2.2.3.1 Topic "IncentiveTable"

3.2.2.3.1.1 Specialization "IncentiveTable_EnergyConsumptionWithTF"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.		
3: M	tariffDescription.		[TOUT-031]
3: M	tariffDescription. tariffId	<tf5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	tariffDescription. tariffWriteable	"true"	[TOUT-035]
3: M	tariffDescription. updateRequired	"true" [TOUT-036]	With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the

		"false"	responsible client. The server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
3: M	tariffDescription. scopeType	"incentiveTableEnConsWithTF"	
3: M	tariffDescription. slotIdSupport	"true"	
3: M \W	tier.		
3: M \W	tier. tierDescription.		
3: M \W	tier. tierDescription. tierId	<tr5#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
3: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
3: R \W	tier. tierDescription. label		[TOUT-004]
3: O \W	tier. tierDescription. description		[TOUT-004]
3: M \W	tier. boundaryDescription.		
3: M \W	tier. boundaryDescription. boundaryId	<by5#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId.
3: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
3: M \W	tier. boundaryDescription. boundaryUnit	"W"	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
3: M \W	tier. incentiveDescription.		
3: M \W	tier. incentiveDescription. incentiveId	<ie5#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId.

3: M \W	tier. incentiveDescription. incentiveType	3: M \W	"absoluteCost "	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as charge. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
3: O \W, M \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: O \W, M \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.			
3: M	tariff.			
3: M	tariff. tariffId	<tf5#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
3: M	tariffConstraints.			
3: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5		[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
3: M	tariffConstraints. maxBoundariesPerTier	≥1		If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
3: M	tariffConstraints. maxIncentivesPerTier	≥1		[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
3: M	incentiveSlotConstraints.			
3: M	incentiveSlotConstraints. slotCountMax			If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
3: M \W	IncentiveTable. incentiveTableData. incentiveTable.			
3: M \W	tariff.			
3: M \W	tariff. tariffId	<tf5#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
3: M \W	incentiveSlot.			
3: M \W	incentiveSlot. timeInterval.			[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.

3: M \W	incentiveSlot. timeInterval. timeSlotId	<ti5#(1..)>	SUB IDENTIFIER of tariffId.
3: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
3: M \W	incentiveSlot. timeInterval. startTime. dateTime		
3: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
3: M \W	incentiveSlot. timeInterval. endTime. dateTime		
3: M \W	incentiveSlot. tier.		
3: M \W	incentiveSlot. tier. tier.		
3: M \W	incentiveSlot. tier. tier. tierId	<tr5#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
3: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
3: M \W	incentiveSlot. tier. boundary. boundaryId	<by5#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	≥0	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	> incentiveSlot. tier. boundary. lowerBoundaryValue	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		

3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
3: M \W	incentiveSlot. tier. incentive.		
3: M \W	incentiveSlot. tier. incentive. incentiveId	<ie5#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId.
3: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
3: M \W	incentiveSlot. tier. incentive. value. number		
3: M \W	incentiveSlot. tier. incentive. value. scale		

Table 15: Content of Specialization "IncentiveTable_EnergyConsumptionWithTF" at Actor Transmission Broker

3.2.2.3.1.2 Specialization "IncentiveTable_EnergyProductionWithTF"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
3: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.		
3: M	tariffDescription.		[TOUT-032]
3: M	tariffDescription. tariffId	<tf6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	tariffDescription. tariffWriteable	"true"	[TOUT-035]
3: M	tariffDescription. updateRequired	"true" [TOUT-036]	With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the

		"false"	responsible client. The server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
3: M	tariffDescription. scopeType	"incentiveTableEnProdWithTF"	
3: M	tariffDescription. slotIdSupport	"true"	
3: M \W	tier.		
3: M \W	tier. tierDescription.		
3: M \W	tier. tierDescription. tierId	<tr6#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
3: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
3: R \W	tier. tierDescription. label		[TOUT-004]
3: O \W	tier. tierDescription. description		[TOUT-004]
3: M \W	tier. boundaryDescription.		
3: M \W	tier. boundaryDescription. boundaryId	<by6#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId.
3: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
3: M \W	tier. boundaryDescription. boundaryUnit	"W"	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
3: M \W	tier. incentiveDescription.		
3: M \W	tier. incentiveDescription. incentiveId	<ie6#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId.

3: M \W	tier. incentiveDescription. incentiveType	3: M \W	"absoluteCost "	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as charge. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
3: O \W, M \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: O \W, M \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.			
3: M	tariff.			
3: M	tariff. tariffId	<tf6#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
3: M	tariffConstraints.			
3: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5		[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
3: M	tariffConstraints. maxBoundariesPerTier	≥1		If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
3: M	tariffConstraints. maxIncentivesPerTier	≥1		[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
3: M	incentiveSlotConstraints.			
3: M	incentiveSlotConstraints. slotCountMax			If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
3: M \W	IncentiveTable. incentiveTableData. incentiveTable.			
3: M \W	tariff.			
3: M \W	tariff. tariffId	<tf6#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
3: M \W	incentiveSlot.			
3: M \W	incentiveSlot. timeInterval.			[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.

3: M \W	incentiveSlot. timeInterval. timeSlotId	<ti6#(1..)>	SUB IDENTIFIER of tariffId.
3: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
3: M \W	incentiveSlot. timeInterval. startTime. dateTime		
3: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
3: M \W	incentiveSlot. timeInterval. endTime. dateTime		
3: M \W	incentiveSlot. tier.		
3: M \W	incentiveSlot. tier. tier.		
3: M \W	incentiveSlot. tier. tier. tierId	<tr6#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
3: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
3: M \W	incentiveSlot. tier. boundary. boundaryId	<by6#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	< incentiveSlot. tier. boundary. upperBoundaryValue	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	≤0	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		

3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
3: M \W	incentiveSlot. tier. incentive.		
3: M \W	incentiveSlot. tier. incentive. incentiveId	<ie6#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId.
3: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
3: M \W	incentiveSlot. tier. incentive. value. number		
3: M \W	incentiveSlot. tier. incentive. value. scale		

Table 16: Content of Specialization "IncentiveTable_EnergyProductionWithTF" at Actor Transmission Broker

3.2.3 CEM

3.2.3.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "CEM" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor CEM's resource hierarchy.

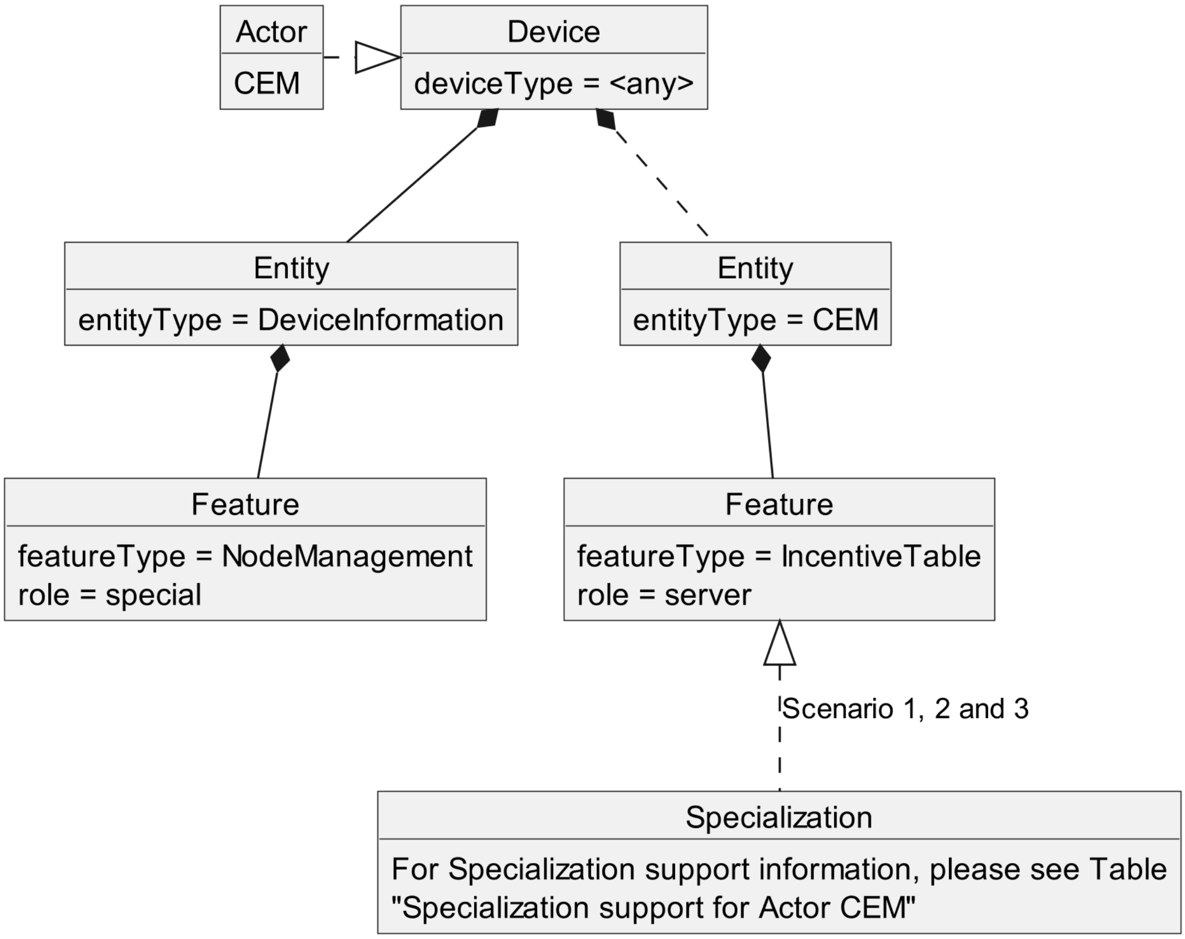

Figure 9: Actor "CEM" overview

The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind the entityType "CEM". The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Used Feature...	...in tables
IncentiveTable_EnergyConsumptionWithPoEandTF	1	IncentiveTable	Table 19
			Table 20
			Table 21
IncentiveTable_EnergyProductionWithPoEandTF	1	IncentiveTable	Table 19
			Table 20
			Table 21
IncentiveTable_EnergyConsumptionWithPoE	2	IncentiveTable	Table 19
			Table 20
			Table 21
IncentiveTable_EnergyProductionWithPoE	2	IncentiveTable	Table 19
			Table 20
			Table 21
IncentiveTable_EnergyConsumptionWithTF	3	IncentiveTable	Table 19
			Table 20
			Table 21
IncentiveTable_EnergyProductionWithTF	3	IncentiveTable	Table 19
			Table 20
			Table 21

Table 17: Specialization support for Actor CEM

3.2.3.2 Server data - Resources

3.2.3.2.1 Overview

Behind the entityType "CEM", the Actor CEM SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
IncentiveTable	1: M 2: M 3: M	incentiveTableDescriptionData	read (M). partial (R) write (M). partial (M)
	1: M 2: M 3: M	incentiveTableConstraintsData	read (M). partial (R) write (M). partial (M)
	1: M 2: M 3: M	incentiveTableData	read (M). partial (R) write (M). partial (M)

Table 18: Feature Types and Functions used within this Use Case by the Actor CEM

For each of these Feature Types the following rule applies: There SHALL be at maximum one Feature with the Feature Type in the Entity.

Note: As a consequence of the previous rule, an implementation may need to have Feature data from different Scenarios/Specializations or even Use Cases in a given Feature.

The Scenario number shows in which Scenarios the CEM acts as server and which Feature Types and Functions are relevant in each Scenario.

A detailed definition of the Elements and values that shall be supported in each Function is given in the following sub-sections.

Note: If in the table above "partial" read is not mentioned or is only optional, it still might be mandatory to support partial notifications. The details of "partial" support are described within the Scenario sections.

Note: The presence indications stated above are meant relative to the ones of the according Scenario stated in Table 1. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the Feature Type need only be supported if the Scenario is supported, too.

Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be implemented in the used Entities.

3.2.3.2.2 Feature Type "IncentiveTable"

3.2.3.2.2.1 Function "incentiveTableDescriptionData"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.		
1: M	tariffDescription.		[TOUT-011]
1: M	tariffDescription. tariffId	<tf1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	tariffDescription. tariffWriteable	"true"	[TOUT-015]
1: M	tariffDescription. updateRequired	"true" [TOUT-016]	With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The

		"false"	server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
1: M	tariffDescription. scopeType	"incentiveTableEnCons WithPoETF"	
1: M	tariffDescription. slotIdSupport	"true"	
1: M \W	tier.		
1: M \W	tier. tierDescription.		
1: M \W	tier. tierDescription. tierId	<tr1#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
1: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
1: R \W	tier. tierDescription. label		[TOUT-004]
1: O \W	tier. tierDescription. description		[TOUT-004]
1: M \W	tier. boundaryDescription.		
1: M \W	tier. boundaryDescription. boundaryId	<by1#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId, if the tier contains more than one boundary entry. If the boundaryId is omitted, a default value of "1" SHALL be applied.
1: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
1: M \W	tier. boundaryDescription. boundaryUnit	"W"	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
1: M \W	tier. incentiveDescription.		

1: M \W	tier. incentiveDescription. incentiveId	<ie1#(1..3)>		SHALL be set as SUB IDENTIFIER of tierId.
1: M \W	tier. incentiveDescription. incentiveType	1: M \W	"absoluteCost"	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as costs. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
		1: O \W	"renewableEnergyPercentage"	[TOUT-007] The "unit" Element of the incentive SHALL be set to "pct" (percentage). The value SHALL be in the range from 0% to 100%.
		1: O \W	"co2Emission"	[TOUT-008] The "unit" Element of the incentive SHALL be set to "kg/Wh". Only positive values SHALL be used.
1: O \W, M \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
1: O \W, M \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
1: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.			
1: M	tariffDescription.			[TOUT-012]
1: M	tariffDescription. tariffId	<tf2#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
1: M	tariffDescription. tariffWriteable	"true"		[TOUT-015]
1: M	tariffDescription. updateRequired	"true" [TOUT-016]		With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The

		"false"	server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
1: M	tariffDescription. scopeType	"incentiveTableEnProd WithPoETF"	
1: M	tariffDescription. slotIdSupport	"true"	
1: M \W	tier.		
1: M \W	tier. tierDescription.		
1: M \W	tier. tierDescription. tierId	<tr2#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
1: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
1: R \W	tier. tierDescription. label		[TOUT-004]
1: O \W	tier. tierDescription. description		[TOUT-004]
1: M \W	tier. boundaryDescription.		
1: M \W	tier. boundaryDescription. boundaryId	<by2#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId, if the tier contains more than one boundary entry. If the boundaryId is omitted, a default value of "1" SHALL be applied.
1: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
1: M \W	tier. boundaryDescription. boundaryUnit	"W"	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
1: M \W	tier. incentiveDescription.		

1: M \W	tier. incentiveDescription. incentiveId	<ie2#(1..1)>		SHALL be set as SUB IDENTIFIER of tierId.
1: M \W	tier. incentiveDescription. incentiveType	1: M \W	"absoluteCost"	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as costs. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
1: O \W, M \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
1: O \W, M \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.			
2: M	tariffDescription.			[TOUT-021]
2: M	tariffDescription. tariffId	<tf3#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
2: M	tariffDescription. tariffWriteable	"true"		[TOUT-025]
2: M	tariffDescription. updateRequired	"true" [TOUT-026]		With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The

		"false"	server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
2: M	tariffDescription. scopeType	"incentiveTableEnConsWithPoE"	
2: M	tariffDescription. slotIdSupport	"true"	
2: M \W	tier.		
2: M \W	tier. tierDescription.		
2: M \W	tier. tierDescription. tierId	<tr3#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
2: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
2: R \W	tier. tierDescription. label		[TOUT-004]
2: O \W	tier. tierDescription. description		[TOUT-004]
2: M \W	tier. boundaryDescription.		
2: M \W	tier. boundaryDescription. boundaryId	<by3#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId, if the tier contains more than one boundary entry. If the boundaryId is omitted, a default value of "1" SHALL be applied.
2: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
2: M \W	tier. boundaryDescription. boundaryUnit	"W"	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
2: M \W	tier. incentiveDescription.		

2: M \W	tier. incentiveDescription. incentiveId	<ie3#(1..3)>		SHALL be set as SUB IDENTIFIER of tierId.
2: M \W	tier. incentiveDescription. incentiveType	2: M \W	"absoluteCost"	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as costs. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
		2: O \W	"renewableEnergyPercentage"	[TOUT-007] The "unit" Element of the incentive SHALL be set to "pct" (percentage). The value SHALL be in the range from 0% to 100%.
		2: O \W	"co2Emission"	[TOUT-008] The "unit" Element of the incentive SHALL be set to "kg/Wh". Only positive values SHALL be used.
2: O \W, M \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: O \W, M \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.			
2: M	tariffDescription.			[TOUT-022]
2: M	tariffDescription. tariffId	<tf4#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
2: M	tariffDescription. tariffWriteable	"true"		[TOUT-025]
2: M	tariffDescription. updateRequired	"true" [TOUT-026]		With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The

		"false"	server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
2: M	tariffDescription. scopeType	"incentiveTableEnProd WithPoE"	
2: M	tariffDescription. slotIdSupport	"true"	
2: M \W	tier.		
2: M \W	tier. tierDescription.		
2: M \W	tier. tierDescription. tierId	<tr4#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
2: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
2: R \W	tier. tierDescription. label		[TOUT-004]
2: O \W	tier. tierDescription. description		[TOUT-004]
2: M \W	tier. boundaryDescription.		
2: M \W	tier. boundaryDescription. boundaryId	<by4#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId, if the tier contains more than one boundary entry. If the boundaryId is omitted, a default value of "1" SHALL be applied.
2: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
2: M \W	tier. boundaryDescription. boundaryUnit	"W"	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
2: M \W	tier. incentiveDescription.		

2: M \W	tier. incentiveDescription. incentiveId	<ie4#(1..1)>		SHALL be set as SUB IDENTIFIER of tierId.
2: M \W	tier. incentiveDescription. incentiveType	2: M \W	"absoluteCost"	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as costs. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
2: O \W, M \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: O \W, M \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.			
3: M	tariffDescription.			[TOUT-031]
3: M	tariffDescription. tariffId	<tf5#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
3: M	tariffDescription. tariffWriteable	"true"		[TOUT-035]
3: M	tariffDescription. updateRequired	"true" [TOUT-036]		With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The

		"false"	server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
3: M	tariffDescription. scopeType	"incentiveTableEnConsWithTF"	
3: M	tariffDescription. slotIdSupport	"true"	
3: M \W	tier.		
3: M \W	tier. tierDescription.		
3: M \W	tier. tierDescription. tierId	<tr5#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
3: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
3: R \W	tier. tierDescription. label		[TOUT-004]
3: O \W	tier. tierDescription. description		[TOUT-004]
3: M \W	tier. boundaryDescription.		
3: M \W	tier. boundaryDescription. boundaryId	<by5#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId, if the tier contains more than one boundary entry. If the boundaryId is omitted, a default value of "1" SHALL be applied.
3: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
3: M \W	tier. boundaryDescription. boundaryUnit	"W"	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
3: M \W	tier. incentiveDescription.		

3: M \W	tier. incentiveDescription. incentiveId	<ie5#(1..1)>		SHALL be set as SUB IDENTIFIER of tierId.
3: M \W	tier. incentiveDescription. incentiveType	3: M \W	"absoluteCost"	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as costs. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
3: O \W, M \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: O \W, M \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.			
3: M	tariffDescription.			[TOUT-032]
3: M	tariffDescription. tariffId	<tf6#(1..1)>		SHALL be set as PRIMARY IDENTIFIER.
3: M	tariffDescription. tariffWriteable	"true"		[TOUT-035]
3: M	tariffDescription. updateRequired	"true" [TOUT-036]		With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is able to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The

		"false"	server SHALL set the updateRequired back to "false", as soon as "incentiveTableDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false". The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
3: M	tariffDescription. scopeType	"incentiveTableEnProd WithTF"	
3: M	tariffDescription. slotIdSupport	"true"	
3: M \W	tier.		
3: M \W	tier. tierDescription.		
3: M \W	tier. tierDescription. tierId	<tr6#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
3: M \W	tier. tierDescription. tierType	"dynamicCost"	The tier has a cost incentive that MAY vary over time.
3: R \W	tier. tierDescription. label		[TOUT-004]
3: O \W	tier. tierDescription. description		[TOUT-004]
3: M \W	tier. boundaryDescription.		
3: M \W	tier. boundaryDescription. boundaryId	<by6#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId, if the tier contains more than one boundary entry. If the boundaryId is omitted, a default value of "1" SHALL be applied.
3: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	The boundary SHALL be an electrical power. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
3: M \W	tier. boundaryDescription. boundaryUnit	"W"	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
3: M \W	tier. incentiveDescription.		

3: M \W	tier. incentiveDescription. incentiveId	<ie6#(1..1)>		SHALL be set as SUB IDENTIFIER of tierId.
3: M \W	tier. incentiveDescription. incentiveType	3: M \W	"absoluteCost"	[TOUT-006] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as costs. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
3: O \W, M \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: O \W, M \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.

Table 19: Content of Function "incentiveTableDescriptionData" at Actor CEM

3.2.3.2.2.2 Function "incentiveTableConstraintsData"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.		
1: M	tariff.		
1: M	tariff. tariffId	<tf1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	tariffConstraints.		
1: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5	[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
1: M	tariffConstraints. maxBoundariesPerTier	≥1	If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
1: M	tariffConstraints. maxIncentivesPerTier	≥1	[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
1: M	incentiveSlotConstraints.		

1: M	incentiveSlotConstraints. slotCountMax		If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
1: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.		
1: M	tariff.		
1: M	tariff. tariffId	<tf2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	tariffConstraints.		
1: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5	[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
1: M	tariffConstraints. maxBoundariesPerTier	≥1	If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
1: M	tariffConstraints. maxIncentivesPerTier	≥1	[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
1: M	incentiveSlotConstraints.		
1: M	incentiveSlotConstraints. slotCountMax		If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
2: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.		
2: M	tariff.		
2: M	tariff. tariffId	<tf3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	tariffConstraints.		
2: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5	[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
2: M	tariffConstraints. maxBoundariesPerTier	≥1	If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
2: M	tariffConstraints. maxIncentivesPerTier	≥1	[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
2: M	incentiveSlotConstraints.		

2: M	incentiveSlotConstraints. slotCountMax		If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
2: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.		
2: M	tariff.		
2: M	tariff. tariffId	<tf4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	tariffConstraints.		
2: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5	[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
2: M	tariffConstraints. maxBoundariesPerTier	≥1	If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
2: M	tariffConstraints. maxIncentivesPerTier	≥1	[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
2: M	incentiveSlotConstraints.		
2: M	incentiveSlotConstraints. slotCountMax		If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
3: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.		
3: M	tariff.		
3: M	tariff. tariffId	<tf5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	tariffConstraints.		
3: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5	[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
3: M	tariffConstraints. maxBoundariesPerTier	≥1	If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
3: M	tariffConstraints. maxIncentivesPerTier	≥1	[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
3: M	incentiveSlotConstraints.		

3: M	incentiveSlotConstraints. slotCountMax		If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
3: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.		
3: M	tariff.		
3: M	tariff. tariffId	<tf6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	tariffConstraints.		
3: M	tariffConstraints. maxTiersPerTariff	M: ≥3 R: ≥5	[TOUT-003] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
3: M	tariffConstraints. maxBoundariesPerTier	≥1	If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
3: M	tariffConstraints. maxIncentivesPerTier	≥1	[TOUT-005] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
3: M	incentiveSlotConstraints.		
3: M	incentiveSlotConstraints. slotCountMax		If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.

Table 20: Content of Function "incentiveTableConstraintsData" at Actor CEM

3.2.3.2.2.3 Function "incentiveTableData"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
1: M \W	tariff.		
1: M \W	tariff. tariffId	<tf1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M \W	incentiveSlot.		
1: M \W	incentiveSlot. timeInterval.		[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
1: M \W	incentiveSlot. timeInterval. timeSlotId	<ti1#(1..)>	SUB IDENTIFIER of tariffId.

1: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
1: M \W	incentiveSlot. timeInterval. startTime. dateTime		
1: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
1: M \W	incentiveSlot. timeInterval. endTime. dateTime		
1: M \W	incentiveSlot. tier.		
1: M \W	incentiveSlot. tier. tier.		
1: M \W	incentiveSlot. tier. tier. tierId	<tr1#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
1: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
1: M \W	incentiveSlot. tier. boundary. boundaryId	<by1#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	≥0	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	> incentiveSlot. tier. boundary. lowerBoundaryValue	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		

1: M \W	incentiveSlot. tier. incentive.		
1: M \W	incentiveSlot. tier. incentive. incentiveId	<ie1#(1..3)>	SHALL be used as SUB IDENTIFIER of tierId.
1: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
1: M \W	incentiveSlot. tier. incentive. value. number		
1: M \W	incentiveSlot. tier. incentive. value. scale		
1: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
1: M \W	tariff.		
1: M \W	tariff. tariffId	<tf2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M \W	incentiveSlot.		
1: M \W	incentiveSlot. timeInterval.		[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
1: M \W	incentiveSlot. timeInterval. timeSlotId	<ti2#(1..)>	SUB IDENTIFIER of tariffId.
1: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
1: M \W	incentiveSlot. timeInterval. startTime. dateTime		
1: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
1: M \W	incentiveSlot. timeInterval. endTime. dateTime		
1: M \W	incentiveSlot. tier.		
1: M \W	incentiveSlot. tier. tier.		
1: M \W	incentiveSlot. tier. tier. tierId	<tr2#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
1: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.

1: M \W	incentiveSlot. tier. boundary. boundaryId	<by2#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	< incentiveSlot. tier. boundary. upperBoundaryValue	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
1: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	≤0	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		
1: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
1: M \W	incentiveSlot. tier. incentive.		
1: M \W	incentiveSlot. tier. incentive. incentiveId	<ie2#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId.
1: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
1: M \W	incentiveSlot. tier. incentive. value. number		
1: M \W	incentiveSlot. tier. incentive. value. scale		
2: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
2: M \W	tariff.		
2: M \W	tariff. tariffId	<tf3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M \W	incentiveSlot.		
2: M \W	incentiveSlot. timeInterval.		[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
2: M \W	incentiveSlot. timeInterval. timeSlotId	<ti3#(1..)>	SUB IDENTIFIER of tariffId.
2: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.

2: M \W	incentiveSlot. timeInterval. startTime. dateTime		
2: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
2: M \W	incentiveSlot. timeInterval. endTime. dateTime		
2: M \W	incentiveSlot. tier.		
2: M \W	incentiveSlot. tier. tier.		
2: M \W	incentiveSlot. tier. tier. tierId	<tr3#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
2: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
2: M \W	incentiveSlot. tier. boundary. boundaryId	<by3#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	≥0	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	> incentiveSlot. tier. boundary. lowerBoundaryValue	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
2: M \W	incentiveSlot. tier. incentive.		

2: M \W	incentiveSlot. tier. incentive. incentiveId	<ie3#(1..3)>	SHALL be used as SUB IDENTIFIER of tierId.
2: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
2: M \W	incentiveSlot. tier. incentive. value. number		
2: M \W	incentiveSlot. tier. incentive. value. scale		
2: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
2: M \W	tariff.		
2: M \W	tariff. tariffId	<tf4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M \W	incentiveSlot.		
2: M \W	incentiveSlot. timeInterval.		[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
2: M \W	incentiveSlot. timeInterval. timeSlotId	<ti4#(1..)>	SUB IDENTIFIER of tariffId.
2: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
2: M \W	incentiveSlot. timeInterval. startTime. dateTime		
2: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
2: M \W	incentiveSlot. timeInterval. endTime. dateTime		
2: M \W	incentiveSlot. tier.		
2: M \W	incentiveSlot. tier. tier.		
2: M \W	incentiveSlot. tier. tier. tierId	<tr4#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
2: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
2: M \W	incentiveSlot. tier. boundary. boundaryId	<by4#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.

2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	< incentiveSlot. tier. boundary. upperBoundaryValue	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
2: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	≤0	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		
2: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
2: M \W	incentiveSlot. tier. incentive.		
2: M \W	incentiveSlot. tier. incentive. incentiveId	<ie4#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId.
2: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
2: M \W	incentiveSlot. tier. incentive. value. number		
2: M \W	incentiveSlot. tier. incentive. value. scale		
3: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
3: M \W	tariff.		
3: M \W	tariff. tariffId	<tf5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M \W	incentiveSlot.		
3: M \W	incentiveSlot. timeInterval.		[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
3: M \W	incentiveSlot. timeInterval. timeSlotId	<ti5#(1..)>	SUB IDENTIFIER of tariffId.
3: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
3: M \W	incentiveSlot. timeInterval. startTime. dateTime		

3: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
3: M \W	incentiveSlot. timeInterval. endTime. dateTime		
3: M \W	incentiveSlot. tier.		
3: M \W	incentiveSlot. tier. tier.		
3: M \W	incentiveSlot. tier. tier. tierId	<tr5#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
3: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
3: M \W	incentiveSlot. tier. boundary. boundaryId	<by5#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	≥0	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	> incentiveSlot. tier. boundary. lowerBoundaryValue	[TOUT-001a], [TOUT-002a], [TOUT-002c], [TOUT-002d] [Scaled number rules]
3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		
3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
3: M \W	incentiveSlot. tier. incentive.		
3: M \W	incentiveSlot. tier. incentive. incentiveId	<ie5#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId.
3: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]

3: M \W	incentiveSlot. tier. incentive. value. number		
3: M \W	incentiveSlot. tier. incentive. value. scale		
3: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
3: M \W	tariff.		
3: M \W	tariff. tariffId	<tf6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M \W	incentiveSlot.		
3: M \W	incentiveSlot. timeInterval.		[TOUT-009] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
3: M \W	incentiveSlot. timeInterval. timeSlotId	<ti6#(1..)>	SUB IDENTIFIER of tariffId.
3: M \W	incentiveSlot. timeInterval. startTime.		[TOUT-010] SHALL be set for all slots.
3: M \W	incentiveSlot. timeInterval. startTime. dateTime		
3: M \W, O \W	incentiveSlot. timeInterval. endTime.		[TOUT-010] SHALL be set for the last slot. MAY be set for all other slots.
3: M \W	incentiveSlot. timeInterval. endTime. dateTime		
3: M \W	incentiveSlot. tier.		
3: M \W	incentiveSlot. tier. tier.		
3: M \W	incentiveSlot. tier. tier. tierId	<tr6#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
3: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
3: M \W	incentiveSlot. tier. boundary. boundaryId	<by6#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.	< incentiveSlot. tier. boundary. upperBoundaryValue	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]

3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue.	≤0	[TOUT-001b], [TOUT-002b], [TOUT-002c], [TOUT-002d] [Scaled number rules]
3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. number		
3: M \W	incentiveSlot. tier. boundary. upperBoundaryValue. scale		
3: M \W	incentiveSlot. tier. incentive.		
3: M \W	incentiveSlot. tier. incentive. incentiveId	<ie6#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId.
3: M \W	incentiveSlot. tier. incentive. value.		[Scaled number rules]
3: M \W	incentiveSlot. tier. incentive. value. number		
3: M \W	incentiveSlot. tier. incentive. value. scale		

Table 21: Content of Function "incentiveTableData" at Actor CEM

3.2.3.3 Client data - Specializations

As this Actor has only server functionality, no Specializations are described within this section.

3.3 Pre-Scenario communication

3.3.1 General information

The Pre-Scenario communication is needed if a client does not know the corresponding addresses on the server or if the required subscriptions or bindings are not active. In this case certain general communication mechanisms SHALL be used within SPINE:

a) Detailed discovery: allows to discover resource addresses.
b) Binding: allows to bind to resource address, which is frequently necessary to obtain write privileges.
c) Subscription: allows to subscribe to resource addresses, which is necessary to receive unsolicited notifications if a resource changes during runtime.

It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription needs to occur.
- E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs to occur only once for all Use Cases.

Depending on which Entity, Feature and Functions are used within a Scenario the payload of the corresponding messages may slightly differ, but the basic principles and messages used stay the same.

The subsequent messages SHALL be exchanged for those parts that have not already been performed since the current connection is established or if those parts are outdated for another reason (e.g. if the detailed discovery is needed, but the bindings and subscriptions are still active from a previous connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario communication parts are up to date, this section MAY be skipped, and the implementation can proceed as described in the corresponding "Scenario communication" sections.

After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-Scenario communication SHALL be performed as well. There may be circumstances where messages from the Pre-Scenario communication may be exchanged again.

Often the necessary messages of different Scenarios can be combined, so that only one single message is needed instead of multiple messages for the different Scenarios. This also is the case for the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery is necessary, as well as only one subscription request or binding request is needed for each Feature. Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or binding is necessary.

3.3.2 Detailed discovery

For the functionality where a client already has current detailed discovery information (i.e. independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

Otherwise, the following procedure SHALL be performed in the given order:

1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL acquire a subscription according to the figure provided within this sub-section.
2. A client SHALL read the detailed discovery information according to the figure provided within this sub-section. It SHALL keep the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case that are needed by the client. This means that a client may choose how to store the necessary information. E.g. a client Actor can store the information how to address the necessary Features of the implemented Scenarios but may discard the Entity information.
3. If and as long as a client has a subscription to the detailed discovery information of an Actor and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed discovery information) the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case.

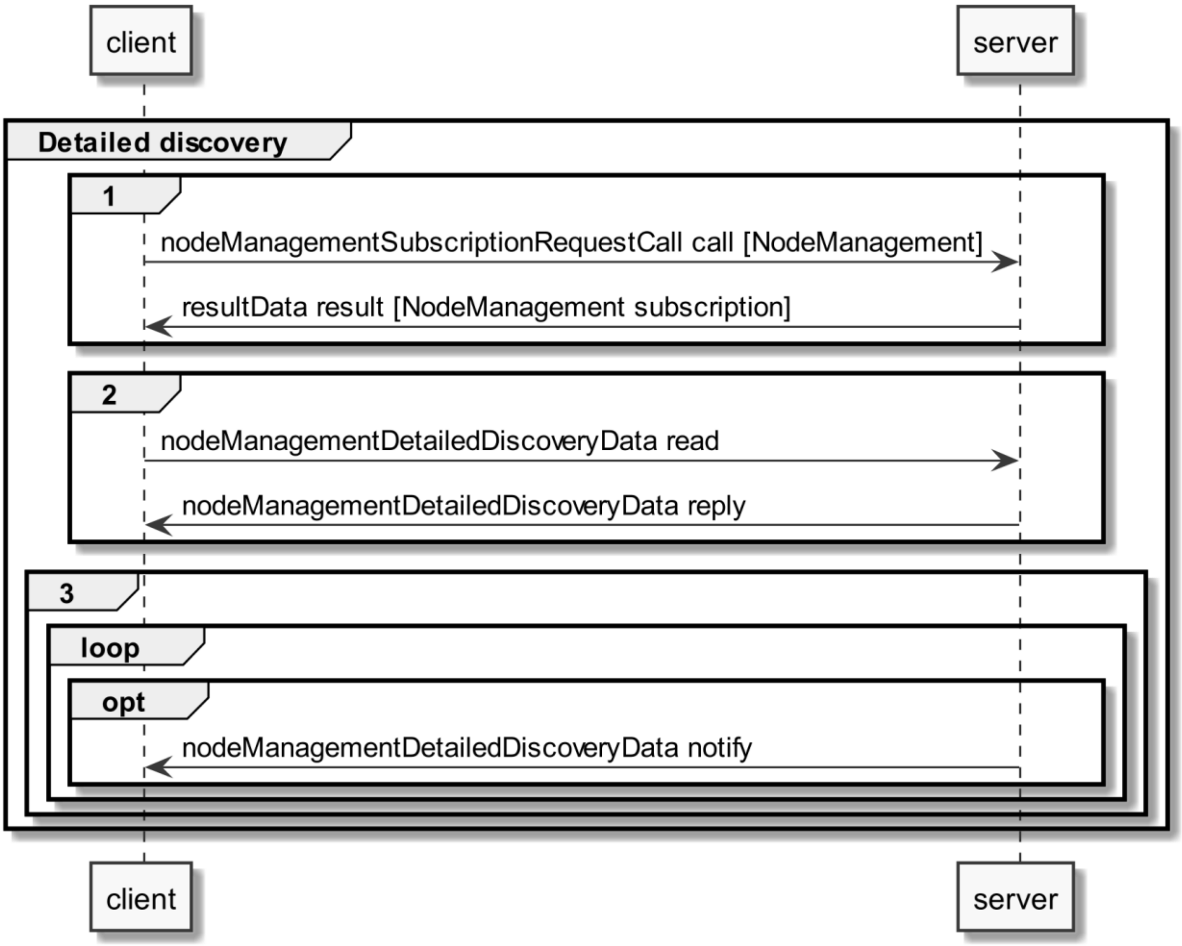

Figure 10: Pre-Scenario communication - Detailed discovery sequence diagram

If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message was received successfully.

If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply" message contains at least the mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as the used Functions and their "possible operations" described in section 3.2 and its sub-sections.

If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as all needed Functions and their "possible operations" described in section 3.2 and its sub-sections. However, as soon as the functionality is available it will be announced via a "nodeManagementDetailedDiscoveryData notify" message.

For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial read with separated Selectors that may use one of the following Elements:

- entityType
- featureType

Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case may be discovered. However, only Features and Entities that fulfil the hierarchical order as described within the Actors' sections shall be considered for this Use Case.

A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify SHOULD also be supported with separated Selectors that may use one of the following Elements:

- entityAddress
- featureAddress

3.3.3 Binding

If binding is required by a Scenario that uses Features with writeable or changeable data, the server SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs, the client SHALL create a binding to the corresponding Feature. For this the nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:

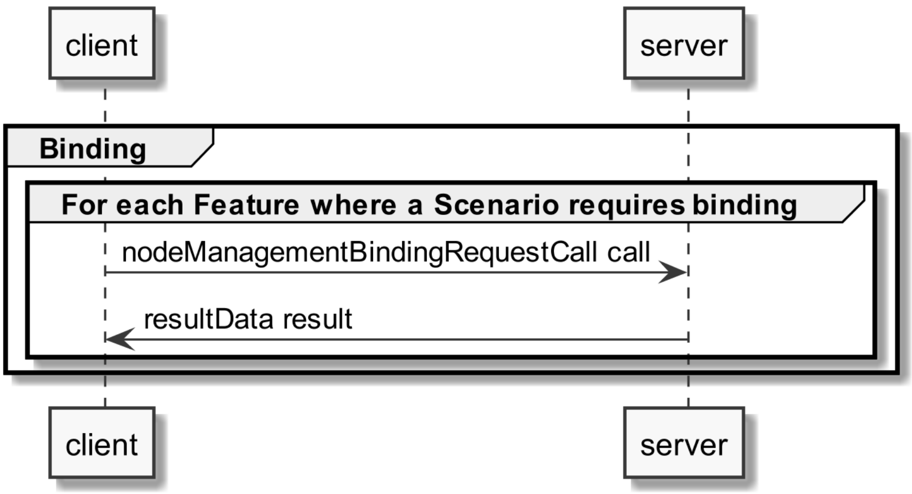

Figure 11: Pre-Scenario communication - Binding sequence diagram

If functionality is added or removed dynamically, binding may not be possible at all times on the required Functions. A client SHALL retry to create a binding again when receiving according updated detailed discovery information.

3.3.4 Subscription

A server SHALL support subscription for all Features that contain readable data that may change during runtime. The client SHALL create a subscription for all Features that the client wants to read. For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following sequence diagram:

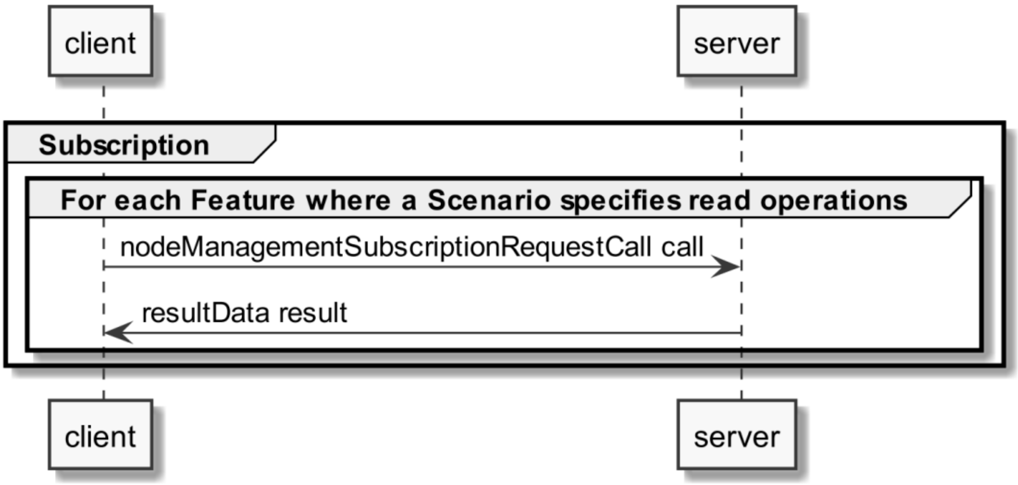

Figure 12: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client SHOULD read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client SHALL retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features SHALL be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

3.4 Scenarios

3.4.1 Scenario 1 - Energy Broker sends incentive table with PoE and TF

3.4.1.1 Pre-Scenario communication

1. **Detailed discovery**: Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding**: Actors that write parts of a Feature within this Scenario, need to create a binding, as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.1.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

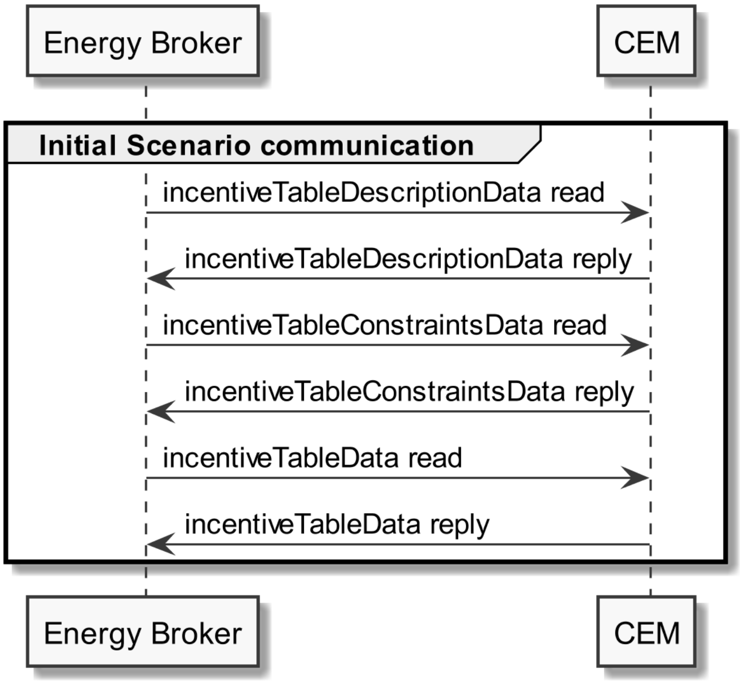

Figure 13: Scenario 1 - Initial Scenario communication sequence diagram

Partial data information regarding [TOUT-011]:

The incentiveTableDescriptionData read SHOULD be a "partial" read operation with the following Selector:

- tariffDescription.scopeType = "incentiveTableEnConsWithPoETF"

The incentiveTableConstraintsData read and incentiveTableData read SHOULD be "partial" read operation with the following Selector:

- tariff.tariffId (derived from the incentiveTableDescriptionData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [TOUT-012]:

The incentiveTableDescriptionData read SHOULD be a "partial" read operation with the following Selector:

- tariffDescription.scopeType = "incentiveTableEnProdWithPoETF"

The incentiveTableConstraintsData read and incentiveTableData read SHOULD be "partial" read operation with the following Selector:

- tariff.tariffId (derived from the incentiveTableDescriptionData reply)

Note: If partial read is not supported, a full read SHALL be performed.

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	**Content description in table**	**Scenario number in table**
incentiveTableDescriptionData reply	Table 19	1
incentiveTableConstraintsData reply	Table 20	1
incentiveTableData reply	Table 21	1

Table 22: Initial Scenario communication content references for Scenario 1

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.1.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

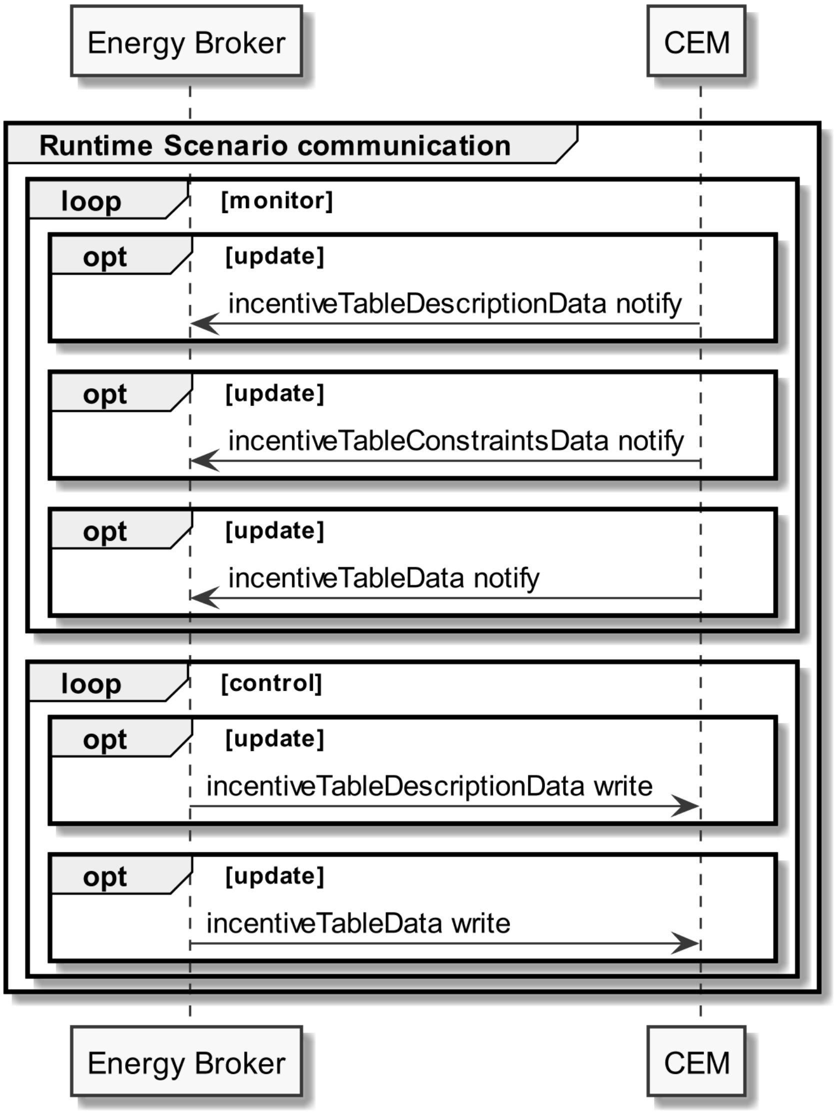

Figure 14: Scenario 1 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the Functions change after a write command by the client during runtime. Hence, usually no notifications of changes by other instances are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For incentiveTableDescriptionData notify "partial" delete notifications SHOULD be supported with the Selector:

- tariffDescription.tariffId

For incentiveTableConstraintsData notify and incentiveTableData notify "partial" delete notifications SHOULD be supported with the Selector:

- tariff.tariffId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

For partial delete write operations the same rules apply as for notifications.

For required content of write messages please consider the respective Specializations.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	**Content description in table**	**Scenario number in table**
incentiveTableDescriptionData notify	Table 19	1
incentiveTableConstraintsData notify	Table 20	1
incentiveTableData notify	Table 21	1
incentiveTableDescriptionData write	Table 19	1
incentiveTableData write	Table 21	1

Table 23: Runtime Scenario communication content references for Scenario 1

3.4.1.4 Additional information

3.4.1.4.1 Rules for Incentive Tables

3.4.1.4.1.1 Recommendations on "timeSlotId"

This Use Case does not prescribe particular values for the Element "timeSlotId". But once a slot (with a dedicated time period) has been added to an Incentive Table, the respective value of "timeSlotId" SHOULD be preserved. Of course, this recommendation applies only as long as an update of an Incentive Table still uses the same time periods for its slots as before.

3.4.1.4.1.2 Recommendations on "partial operations" on slots

This section applies to the Function "incentiveTableData" only. In general, an update of an Incentive Table's slots may comprise of the removal of (past) slots, the addition of new slots, the change of an existing slot, or even a replacement of existing slots if a given period needs to be split into a different number of slots with altered durations. These changes may be expressed with "partial operations" in different ways. As a proper modelling and evaluation of partial changes can be prone to errors, some definitions and recommendations will be given. For details and technical background information please consider [ProtocolSpecification].

"New slots":

1. The Actor Energy Broker MAY append one or more future slots to an existing Incentive Table (without time gaps).

"Past slots":

The following rules aim at cleaning up the Incentive Table to focus on current and subsequent activities:

1. The Actor Energy Broker SHOULD NOT write slots whose end time is in the past.
2. The Actor CEM MAY remove slots whose end time is in the past. This removal SHALL be notified properly.

Partial write operations by Actor Energy Broker to adjust slots:

If the Actor Energy Broker just wants to add "new slots" as described above, the Actor Energy Broker can send a simple "write" operation with a single "cmdControl" Element containing the Element "partial" together with the new slots. This kind of message is also applicable if an already existing slot just need to get a value change in one of its Elements.

In every other case of slot adjustment (remove "past slots", replace existing slots with new slots, etc.) it is recommended to send the "write" operation using a so-called "delete-add" pattern: For a given value of "tariffId" the "write" operation contains a "filter" Element with "delete" as well as a "filter" Element with "partial". This is shown in the following excerpt of a "datagram":

```
...
<payload>
    <cmd>
        <function>incentiveTableData</function>
        <filter>
            <cmdControl><delete/></cmdControl>
            <incentiveTableDataSelectors>
                <tariff>
                    <tariffId>(*1)</tariffId>
                </tariff>
            </incentiveTableDataSelectors>
        </filter>
        <filter>
            <cmdControl><partial/></cmdControl>
        </filter>
        <incentiveTableData>
            (*2)
        </incentiveTableData>
    </cmd>
</payload>
```

where "(*1)" must be replaced by the proper value of "tariffId" and "(*2)" must contain a proper "incentiveTable" instance with the same value of "tariffId" and slots to be stored by Actor CEM. This pattern replaces the previous "incentiveTable" instance (for the given value of "tariffId") with the new one ("(*2)").

Partial notifications by Actor CEM on adjusted slots:

If the Actor CEM just wants to report "new slots" as described above, the Actor CEM can send a simple "notify" operation with a single "cmdControl" Element containing the Element "partial" together with the new slots. This kind of message is also applicable if an already existing slot just got a value change in one of its Elements.

In every other case of slot adjustment (removal of "past slots", replacement of existing slots with new slots, etc.) it is recommended to send the "notify" operation using the same "delete-add" pattern as described above.

3.4.1.4.1.3 Recommendations on "partial operations" on "incentiveTableDescriptionData"

This section applies to the Function "incentiveTableDescriptionData" only. Similar to section 3.4.1.4.1.2, an update of an Incentive Table's incentiveTableDescriptionData may comprise of the removal or addition or change of "tier" instances or other changes. Esp. the Actor Energy Broker needs to consider that the Actor CEM may contain "unexpected" "tier" instances: This can happen if the Energy Broker has been reset to factory default and does not remember previously written tiers. An Energy Broker may also need to remove "tier" instances it previously wrote, but that are not valid anymore due to a switch to a different electricity provider with a new tariff structure, e.g.

In every case, the Actor Energy Broker needs to ensure that no "unknown" or "old" "tier" information is present in the Actor CEM's Incentive Table after the Actor Energy Broker's write operation.

Partial write operations by Actor Energy Broker to remove "unknown" or "old" "tier" information:

Whenever the Energy Broker needs to remove "unknown" or "old" "tier" information it is recommended to send the "write" operation using a so-called "delete-add" pattern: For a given value of "tariffId" the "write" operation contains a "filter" Element with "delete" as well as a "filter" Element with "partial". This is shown in the following excerpt of a "datagram":

```
...
<payload>
    <cmd>
        <function>incentiveTableDescriptionData</function>
        <filter>
            <cmdControl><delete/></cmdControl>
            <incentiveTableDescriptionDataSelectors>
                <tariffDescription>
                    <tariffId>(*1)</tariffId>
                </tariffDescription>
            </incentiveTableDescriptionDataSelectors>
        </filter>
        <filter>
            <cmdControl><partial/></cmdControl>
        </filter>
        <incentiveTableDescriptionData>
            (*2)
        </incentiveTableDescriptionData>
    </cmd>
</payload>
```


where "(*1)" must be replaced by the proper value of "tariffId" and "(*2)" must contain a proper "incentiveTableDescription" instance with the same value of "tariffId" and tier and further data to be stored by Actor CEM. This pattern replaces the previous "incentiveTableDescription" instance (for the given value of "tariffId") with the new one ("(*2)").

Note: This "delete-add" pattern removes all writeable data of the related "incentiveTableDescription" instance, i.e. not just contained "tier" instances. This means the write operation may need to set such Elements as well with current values.

Partial notifications by Actor CEM:

If the Actor CEM just wants to report "new tiers", the Actor CEM can send a simple "notify" operation with a single "cmdControl" Element containing the Element "partial" together with the new "tier" instances. This kind of message is also applicable if an already existing "tier" just got a value change in one of its Elements.

In case of the removal of "tier" instances, it is recommended to send the "notify" operation using the same "delete-add" pattern as described above.

3.4.2 Scenario 2 - Energy Broker sends incentive table with PoE

3.4.2.1 Pre-Scenario communication

1. **Detailed discovery**: Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding**: Actors that write parts of a Feature within this Scenario, need to create a binding, as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.2.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

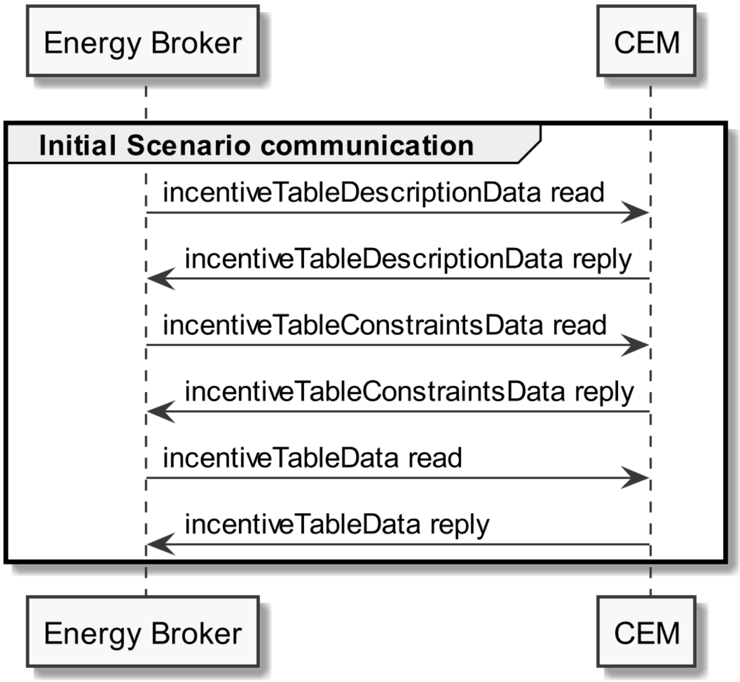

Figure 15: Scenario 2 - Initial Scenario communication sequence diagram

Partial data information regarding [TOUT-021]:

The incentiveTableDescriptionData read SHOULD be a "partial" read operation with the following Selector:

- tariffDescription.scopeType = "incentiveTableEnConsWithPoE"

The incentiveTableConstraintsData read and incentiveTableData read SHOULD be "partial" read operation with the following Selector:

- tariff.tariffId (derived from the incentiveTableDescriptionData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [TOUT-022]:

The incentiveTableDescriptionData read SHOULD be a "partial" read operation with the following Selector:

- tariffDescription.scopeType = "incentiveTableEnProdWithPoE"

The incentiveTableConstraintsData read and incentiveTableData read SHOULD be "partial" read operation with the following Selector:

- tariff.tariffId (derived from the incentiveTableDescriptionData reply)

Note: If partial read is not supported, a full read SHALL be performed.

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
incentiveTableDescriptionData reply	Table 19	2
incentiveTableConstraintsData reply	Table 20	2
incentiveTableData reply	Table 21	2

Table 24: Initial Scenario communication content references for Scenario 2

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.2.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

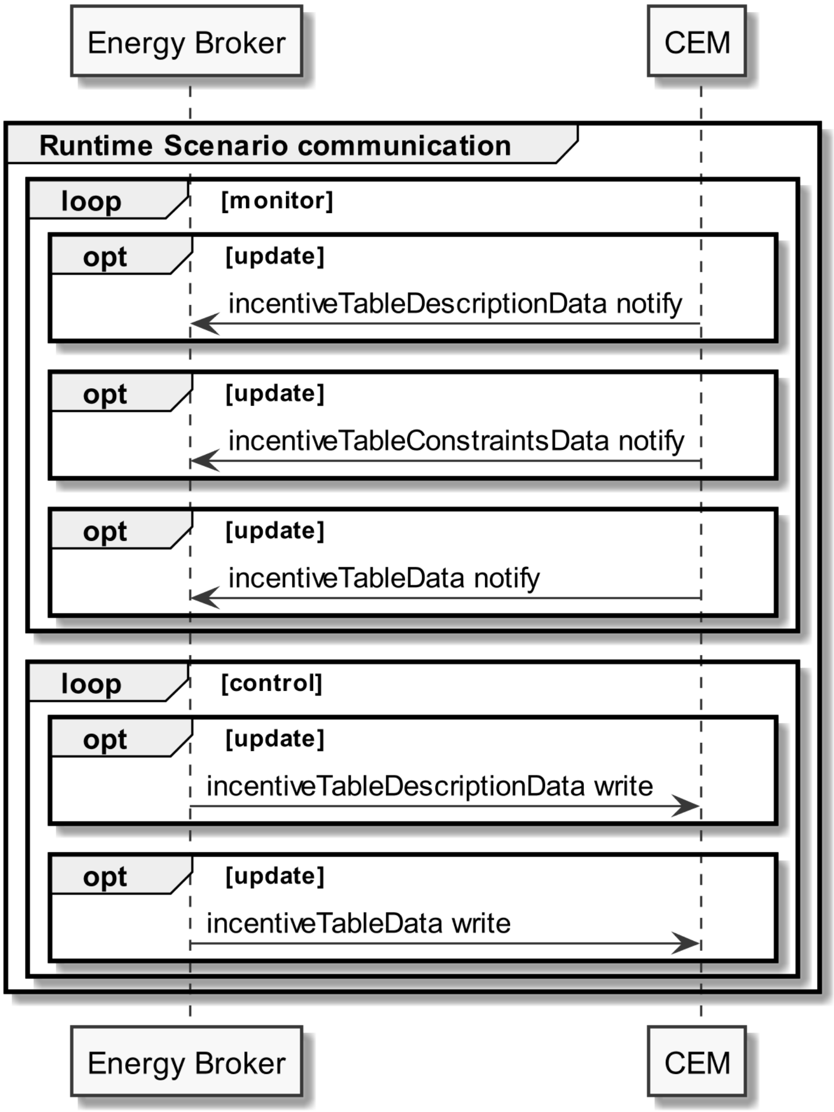

Figure 16: Scenario 2 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the Functions change after a write command by the client during runtime. Hence, usually no notifications of changes by other instances are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For incentiveTableDescriptionData notify "partial" delete notifications SHOULD be supported with the Selector:

- tariffDescription.tariffId

For incentiveTableConstraintsData notify and incentiveTableData notify "partial" delete notifications SHOULD be supported with the Selector:

- tariff.tariffId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

For partial delete write operations the same rules apply as for notifications.

For required content of write messages please consider the respective Specializations.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	**Content description in table**	**Scenario number in table**
incentiveTableDescriptionData notify	Table 19	2
incentiveTableConstraintsData notify	Table 20	2
incentiveTableData notify	Table 21	2
incentiveTableDescriptionData write	Table 19	2
incentiveTableData write	Table 21	2

Table 25: Runtime Scenario communication content references for Scenario 2

3.4.2.4 Additional information

Please refer to section 3.4.1.4 for additional information.

3.4.3 Scenario 3 - Transmission Broker sends incentive table with TF

3.4.3.1 Pre-Scenario communication

1. **Detailed discovery**: Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding**: Actors that write parts of a Feature within this Scenario, need to create a binding, as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.3.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

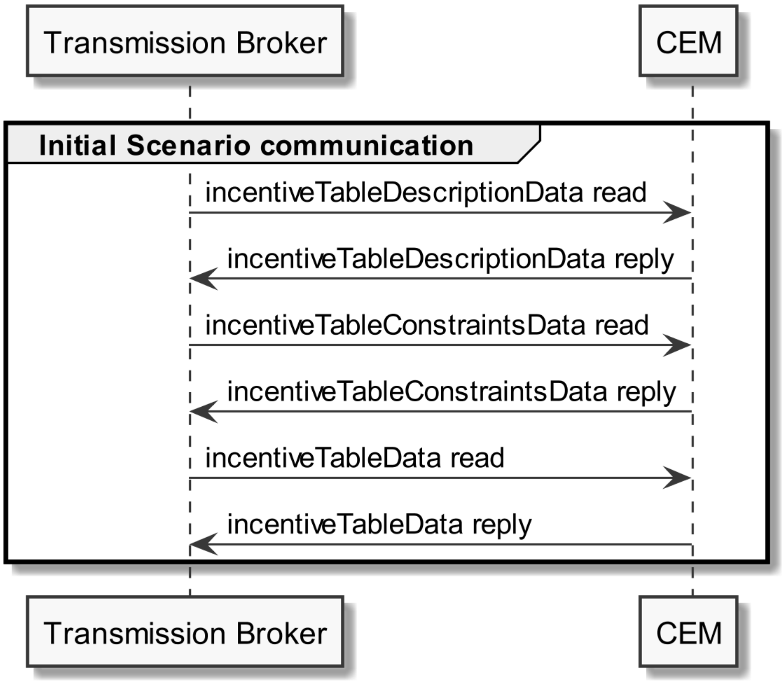

Figure 17: Scenario 3 - Initial Scenario communication sequence diagram

Partial data information regarding [TOUT-031]:

The incentiveTableDescriptionData read SHOULD be a "partial" read operation with the following Selector:

- tariffDescription.scopeType = "incentiveTableEnConsWithTF"

The incentiveTableConstraintsData read and incentiveTableData read SHOULD be "partial" read operation with the following Selector:

- tariff.tariffId (derived from the incentiveTableDescriptionData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [TOUT-032]:

The incentiveTableDescriptionData read SHOULD be a "partial" read operation with the following Selector:

- tariffDescription.scopeType = "incentiveTableEnProdWithTF"

The incentiveTableConstraintsData read and incentiveTableData read SHOULD be "partial" read operation with the following Selector:

- tariff.tariffId (derived from the incentiveTableDescriptionData reply)

Note: If partial read is not supported, a full read SHALL be performed.

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
incentiveTableDescriptionData reply	Table 19	3
incentiveTableConstraintsData reply	Table 20	3
incentiveTableData reply	Table 21	3

Table 26: Initial Scenario communication content references for Scenario 3

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.3.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

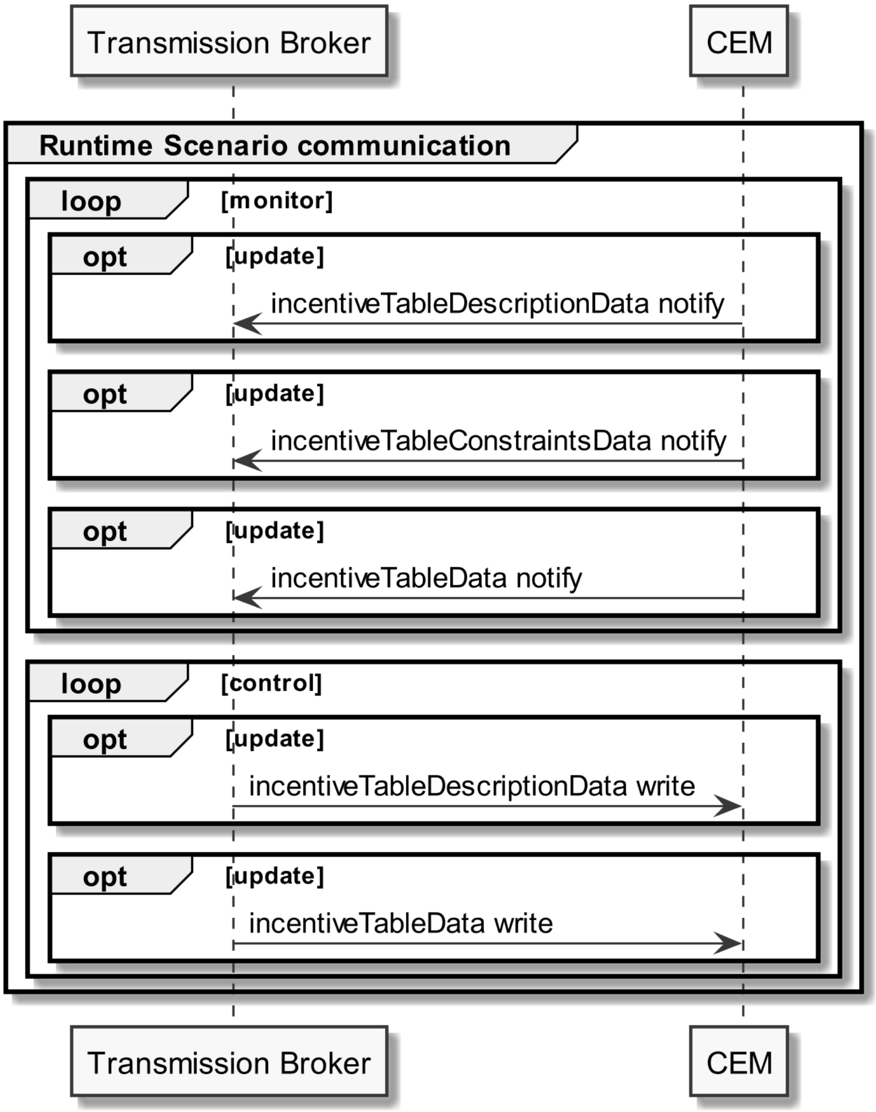

Figure 18: Scenario 3 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the Functions change after a write command by the client during runtime. Hence, usually no notifications of changes by other instances are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For incentiveTableDescriptionData notify "partial" delete notifications SHOULD be supported with the Selector:

- tariffDescription.tariffId

For incentiveTableConstraintsData notify and incentiveTableData notify "partial" delete notifications SHOULD be supported with the Selector:

- tariff.tariffId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

For partial delete write operations the same rules apply as for notifications.

For required content of write messages please consider the respective Specializations.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	**Content description in table**	**Scenario number in table**
incentiveTableDescriptionData notify	Table 19	3
incentiveTableConstraintsData notify	Table 20	3
incentiveTableData notify	Table 21	3
incentiveTableDescriptionData write	Table 19	3
incentiveTableData write	Table 21	3

Table 27: Runtime Scenario communication content references for Scenario 3

3.4.3.4 Additional information

Please refer to section 3.4.1.4 for additional information. The Actor "Energy Broker" in that section needs to be replaced by the Actor "Transmission Broker" for this section.

